# Zero Counts, Heat Transport, and Localized Hodge Energy

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**Abstract**

For a holomorphic function with nonzero boundary values, the derivative-weighted energy over the region $|f| < \varepsilon$ is exactly $4\pi\varepsilon$ times its zero count, provided $\varepsilon$ lies below the boundary amplitude. The formula includes every multiplicity and is the energy of the exact differential $2\,dw$ on the normalized cover $w^2 = f$. We give a general power-energy and root-cover formulation, construct compact target-space regularizations of the divisor, and prove explicit field-error, boundary-bias and localization estimates. Harmonic reproduction gives second-order bias; complete disk and unit control give a weighted-divisor rate $O(\eta^{\min(1,2/m)})$ at an $m$-fold zero, with matching exponent examples. For analytic backward heat flow we prove the complete real cap-and-wall balance with event weight $\lfloor m/2 \rfloor$, distinguish it from conserved complex multiplicity, and identify the complex discriminant residue $\binom{m}{2}$. We then derive finite inverse-square pair-matrix bounds and the complete forced root dynamics. An elementary concentration argument for the full theta source proves clock limits on explicit growing rank windows and fixed-population cumulative exterior cancellation. Finally, weighted pair-law transfer and exact physical cap formulas specify the additional small-gap, analytic-unit and phase information that such data require. The proofs are analytic and finite-domain; a numerical certificate is used only in a supporting appendix. No assertion of RH or endpoint simplicity is made.

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# 1 Introduction

Zeros in an analytic family can be counted in several ways. The argument principle counts complex zeros with multiplicity. A real heat evolution can turn a conjugate pair into two real zeros, and a real spacetime index records that event. A complex-time discriminant counts collisions with yet another multiplicity. On a branched cover, the same zeros determine the topology, while a chosen differential supplies an analytic energy. The usefulness of a boundary balance depends on distinguishing these quantities and retaining the complete boundary terms.

This paper develops a self-contained finite-domain framework linking these descriptions. Its main analytic identity is particularly simple. If $f$ is holomorphic near a bounded domain $\Omega$, has no boundary zero, and has $N$ interior zeros with multiplicity, then

$$2\int_{\Omega\cap\{|f|<\varepsilon\}} \frac{|f'|^2}{|f|}\,dA = 4\pi\varepsilon N, \qquad 0<\varepsilon<\min_{\partial\Omega}|f|. \tag{1.1}$$

There is no simplicity or separation hypothesis. The integral remains locally finite through every finite multiplicity. On the normalized cover $w^2=f(z)$ it is the energy of the exact differential $2\,dw$ over the low-value region. More generally, the energy of $q\,dw$ on $w^q=f(z)$ is exactly $\pi q^2\varepsilon^{2/q}N$ on

the corresponding region. These are applications of the holomorphic degree/area formula; the point is their quantitative formulation and their interaction with transport, rather than a new version of the argument principle.

The localized energy has a compact target-space potential. This gives a regularized divisor measure of exact total mass $N$, an approximation allowance proportional to the field error divided by the value cutoff, and a quantitative weak-limit bias near roots of arbitrary multiplicity. A family of power potentials connects the double-cover case to higher root covers. We also compare the exact-mass regularization with a smooth logarithmic kernel whose smaller Lipschitz constant comes with an explicit boundary bias. These statements specify the information needed to use an energy integral as a count; they do not infer a count from an uncontrolled normalization. Harmonic reproduction removes the linear localization error for $C^2$ tests. With complete disk, boundary and analytic-unit data, a direct contour estimate gives weighted-divisor stability of order $\eta^{\min(1,2/m)}$ at an $m$-fold zero; monomial and translation examples attain the exponent.

For the backward heat equation

$$H_t = -H_{zz}, \tag{1.2}$$

the complex multiplicity count is conserved when roots do not cross the spatial boundary. Real roots obey a different complete balance: an interior spatial multiplicity $m$ contributes $2\lfloor m/2 \rfloor$ new real roots, with signed wall crossings added to the cap difference. The complex discriminant instead has local vanishing order $\binom{m}{2}$. We prove these weights from the local Hermite expansion and give the complete cap conventions. The normalized double cover has a singular-fiber Euler-characteristic excess $2\sum\lfloor m/2 \rfloor$ over a regular fiber with the same total complex count. These formulas show precisely when transported simple zeros cancel and what remains after that cancellation.

Root geometry provides another analytic interface. Inverse-square pair distances form a graph Laplacian, the Hessian of logarithmic Vandermonde energy. We derive its complete heat evolution, retaining the holomorphic factor containing the omitted roots. A full polynomial heat flow has a strongly convex rescaled Hermite energy. For a selected cluster of an entire function, the exterior factor acts as a force. We give an explicit matrix comparison and a self-contained large-time application to the complete theta source: blocks ending by rank $K = o(L\sqrt{\kappa})$ converge after scaling to their clock matrices with a quantitative error uniform in the block size. A fixed-population cumulative cancellation formula explains why the exterior force cannot be discarded.

Pair correlation is a possible additional input, but it uses a different limiting topology. We compute the clock and sine-kernel inverse-square moments, prove a weighted transfer theorem with its small-gap and tail hypotheses, and give an explicit finite-population error threshold that would exclude repeated points. Finally, the physical projective cap formula shows that the root-derivative phase is additional information: regular simple roots alone do not make the cap integral vanish. Its two imaginary-time faces are evaluated with their orientations.

**Analytic scope and relation to earlier work.** The basic ingredients are classical: the argument principle, the holomorphic change-of-variables formula, Green's identity, Riemann–Hurwitz, Hermite polynomials, and logarithmic root interactions. The real and complex transport identities consolidate earlier work in the accompanying research repository. We provide their proofs here so that the present paper does not require those internal notes. The regularization family, explicit stability and localization estimates, matrix bounds, complete-source concentration estimate and boundary phase formulas are the quantitative statements to be assessed on their own hypotheses. No claim of mathematical priority is made.

The theta field uses the standard normalization in Rodgers–Tao [1]. Our application concerns large positive heat time and the specified growing spatial windows, not a conjectured high-height

pair law. Cumulative dynamical identities retain fixed populations. None of the main analytic results assumes RH or proves RH or endpoint simplicity. The only numerical input appears in Appendix A, as an illustration using a previously certified finite disk. Removing that appendix changes none of the proofs or analytic conclusions.

**Conventions.** Spatial area is $dA = dx\,dy$, and the energy of a differential $\eta$ on a Riemann surface is $(i/2)\int \eta \wedge \bar{\eta}$. Complex time is $\tau = t + is$. Ordered pair measures omit only the label diagonal; coincident locations with different labels remain counted. Every boundary orientation is stated when used. The notation $O_K$ allows constants depending on a fixed finite rank bound $K$; it does not assert uniformity as the population grows.

## 2 Localized energy and holomorphic degree

Throughout this section, $\Omega \subset \mathbb{C}$ is a bounded connected domain whose boundary consists of finitely many disjoint piecewise $C^1$ Jordan curves, and $f$ is holomorphic on a neighborhood of $\overline{\Omega}$. All boundary components have their positive orientation relative to $\Omega$, including the clockwise orientation of inner components. Suppose

$$b := \min_{\partial\Omega} |f| > 0, \qquad N := \sum_{a\in\Omega: f(a)=0} \operatorname{ord}_a f. \tag{2.1}$$

The assumptions exclude the identically zero function. They allow a nonzero constant, in which case $N = 0$. Write $dA = dx\,dy$ and $\Delta = \partial_x^2 + \partial_y^2$.

**Lemma 2.1** (Constant valence over a target disk)**.** *For each $w \in \mathbb{C}$ with $|w| < b$, the equation $f(z) = w$ has exactly $N$ solutions in $\Omega$, counted with multiplicity. Consequently, for every nonnegative measurable function $\rho$ supported in $\{|w| < b\}$,*

$$\int_\Omega \rho(f(z))|f'(z)|^2\,dA_z = N\int_{\mathbb{C}} \rho(w)\,dA_w. \tag{2.2}$$

*Both sides may initially be interpreted as nonnegative extended integrals, with the measure-theoretic convention $0\cdot\infty = 0$.*

*Proof.* The homotopy $f - sw$, $0 \le s \le 1$, is nonzero on every boundary component, since $|f - sw| \ge b - |w| > 0$. The argument principle therefore makes the number of zeros independent of $s$, proving the first assertion. For a nonconstant $f$, its critical points in $\overline{\Omega}$ are finite. Away from their finitely many values, the inverse function theorem gives local inverse branches, and the real Jacobian of a holomorphic map is $|f'|^2$. Apply the ordinary change-of-variables formula on a countable partition of such inverse-branch neighborhoods. The number of branches above each regular value in the target disk is $N$. Critical values have planar measure zero. First proving the assertion for bounded simple functions and then using monotone convergence proves (2.2). For a nonzero constant, both integrands are zero in the relevant target disk, and the conclusion is immediate. □

**Theorem 2.2** (Localized power-energy identity)**.** *For every $p > 0$ and $0 < \varepsilon < b$,*

$$\int_{\Omega\cap\{|f|<\varepsilon\}} |f'|^2|f|^{p-2}\,dA = \frac{2\pi}{p}\varepsilon^p N. \tag{2.3}$$

*In particular the localized derivative-weighted energy*

$$W_{<\varepsilon}(f;\Omega) := 2\int_{\Omega\cap\{|f|<\varepsilon\}} \frac{|f'|^2}{|f|}\,dA \tag{2.4}$$

*satisfies*

$$\boxed{W_{<\varepsilon}(f;\Omega) = 4\pi\varepsilon N.} \tag{2.5}$$

*The integrals are finite at zeros of every finite multiplicity.*

*Proof.* Use Lemma 2.1 with $\rho(w) = |w|^{p-2}\mathbf{1}_{\{|w|<\varepsilon\}}$ and evaluate $2\pi\int_0^\varepsilon r^{p-1}\,dr = 2\pi\varepsilon^p/p$. For direct local verification, write $f(z) = (z-a)^m u(z)$ with $u(a) \neq 0$. Then $|f'|^2|f|^{p-2} = O(|z-a|^{mp-2})$, whose area integral near $a$ is finite because $mp > 0$. Setting $p = 1$ and multiplying by two gives (2.5). □

**Corollary 2.3** (Full energy bound)**.** *With $W(f;\Omega) = 2\int_\Omega |f'|^2/|f|\,dA$, one has*

$$W(f;\Omega) \geq 4\pi bN. \tag{2.6}$$

*More generally the full $p$-energy is at least $2\pi b^p N/p$.*

*Proof.* The full integral dominates the localized one for every $\varepsilon < b$. Let $\varepsilon \uparrow b$ in Theorem 2.2. □

The quantity in (2.5) counts all zeros with their multiplicities; it does not separate a multiple zero from its simple descendants. The identity is a degree/area formula, rather than a new integer-valued period. It is exact because the entire target disk lies below the boundary threshold. No bound on individual root gaps occurs.

**Example 2.4** (Sharpness at every multiplicity)**.** For $f(z) = az^m$ on $|z| < R$, where $a \neq 0$, the boundary threshold is $b = |a|R^m$ and $N = m$. Direct radial integration gives

$$W_{<\varepsilon} = 4\pi m\varepsilon, \qquad W = 4\pi m|a|R^m = 4\pi bN.$$

Thus (2.6) is sharp even at a multiple zero.

**Proposition 2.5** (Energy on a finite root cover)**.** *Let $q \geq 1$ be an integer. Normalize $w^q = f(z)$ over an open neighborhood of $\overline{\Omega}$, and let $X_q$ be the inverse image of $\overline{\Omega}$ on that normalization. The differential $\eta_q = q\,dw$ is holomorphic on $X_q$, including points over multiple zeros. Define its localized energy by*

$$\mathcal{E}_{q,<\varepsilon} := \frac{i}{2}\int_{X_q\cap\{|w|^q<\varepsilon\}} \eta_q \wedge \overline{\eta_q}.$$

*For $0 < \varepsilon < b$,*

$$\boxed{\mathcal{E}_{q,<\varepsilon} = q\int_{\Omega\cap\{|f|<\varepsilon\}} |f'|^2|f|^{2/q-2}\,dA = \pi q^2\varepsilon^{2/q}N.} \tag{2.7}$$

*The cover may have several connected components.*

*Proof.* Away from the zeros of $f$, the projection has $q$ sheets and $q\,dw = f'\,dz/w^{q-1}$. Summing the sheetwise energy gives the first equality. The local normalization at an $m$-fold zero, with $d = \gcd(q,m)$, has $d$ charts of the form $z - a = \xi^{q/d}$, $w = \xi^{m/d}v(\xi)$, with $v$ holomorphic and nonzero. Thus $dw$ is regular, and the omitted branch set has zero area. Theorem 2.2 with $p = 2/q$ proves the second equality. Equivalently, above each sufficiently small regular $w$ the map $w : X_q \to \mathbb{C}$ has $N$ preimages; applying the area formula to $q\,dw$ gives $q^2N$ times the area of the disk of radius $\varepsilon^{1/q}$. □

For $q = 2$, $\eta_2 = f'\,dz/w = 2\,dw$, and $\mathcal{E}_{2,<\varepsilon} = W_{<\varepsilon}$. This is the localized Hodge-energy interpretation of (2.5). The original reciprocal differential $dz/w$ is a different differential; its singularities are examined in Section 5.

# 3 Regularized divisor measures and quantitative stability

The localized identity has a potential-theoretic form that avoids locating the roots inside the integral. This section supplies the complete regularization error and the dependence on the field approximation.

## 3.1 A compact target-space regularization

For $p > 0$ define

$$v_{\varepsilon,p}(r) = \begin{cases} \dfrac{1}{p}(r/\varepsilon)^p, & 0 \le r \le \varepsilon, \\ \log(r/\varepsilon) + \dfrac{1}{p}, & r \ge \varepsilon. \end{cases} \tag{3.1}$$

The two branches and their first derivatives agree at $\varepsilon$. For a holomorphic $f$ that is not identically zero, set

$$\mu_{\varepsilon,p}(f) := \frac{1}{2\pi}\Delta v_{\varepsilon,p}(|f|) = \frac{p}{2\pi\varepsilon^p}|f'|^2|f|^{p-2}\mathbf{1}_{\{|f|<\varepsilon\}}\, dA. \tag{3.2}$$

The Laplacian is distributional. For a constant zero function the left side is zero, and we define its measure accordingly.

**Proposition 3.1** (Exact mass and weak divisor limit)**.** *Formula* (3.2) *defines a positive locally finite measure. There is no singular measure on the level $|f| = \varepsilon$, even when that level contains critical points of $f$, and no point mass at a zero for fixed $\varepsilon > 0$. Under* (2.1)*,*

$$\mu_{\varepsilon,p}(f)(\Omega) = N \quad (0 < \varepsilon < b). \tag{3.3}$$

*For $f$ not identically zero, as $\varepsilon \downarrow 0$,*

$$\mu_{\varepsilon,p}(f) \;\longrightarrow\; \sum_{f(a)=0} \mathrm{ord}_a(f)\,\delta_a \tag{3.4}$$

*weakly on compact subsets of the holomorphy domain.*

*Proof.* At $f \neq 0$, the chain rule for a radial function gives

$$\Delta v(|f|) = \left(v''(|f|) + \frac{v'(|f|)}{|f|}\right)|f'|^2.$$

Inside the target disk this is $p\varepsilon^{-p}|f|^{p-2}|f'|^2$; outside it is zero. Near $|f| = \varepsilon$, $v_{\varepsilon,p}(|f|)$ is $C^{1,1}$, because $f \neq 0$ there and the one-sided second derivatives of $v$ are bounded. Hence no interface measure occurs. Near an $m$-fold zero the density has radial order $r^{mp-2}$ and its integral is $O(r^{mp})$. The normal flux of the potential around a radius-$r$ circle also tends to zero at that rate. Removing small circles, integrating by parts and letting their radii tend to zero verifies the distributional identity without atoms.

Theorem 2.2 gives (3.3). For the weak limit, choose disjoint small disks around the finitely many zeros in a compact neighborhood. Once $\varepsilon$ is below the boundary minimum on each disk, its measure is exactly the multiplicity there. The measure vanishes where $|f| \ge \varepsilon$, so its support contracts to those zeros. Uniform continuity of each compactly supported test function completes the proof. □

**Theorem 3.2** (Field-error bounds)**.** *Let $\psi \in C_c^2(\Omega)$ and let $f, g$ be holomorphic on a neighborhood of its support. Suppose $|f - g| \le \eta$ on that support. For $p \ge 1$,*

$$\left| \int \psi \, d\mu_{\varepsilon,p}(f) - \int \psi \, d\mu_{\varepsilon,p}(g) \right| \le \frac{\eta}{2\pi\varepsilon} \|\Delta\psi\|_{L^1}. \tag{3.5}$$

*For $0 < p < 1$ the corresponding bound is*

$$\left| \int \psi \, d\mu_{\varepsilon,p}(f) - \int \psi \, d\mu_{\varepsilon,p}(g) \right| \le \frac{\eta^p}{2\pi p\varepsilon^p} \|\Delta\psi\|_{L^1}. \tag{3.6}$$

*No root separation, zero-free interior or derivative approximation is assumed.*

*Proof.* The distributional definition gives

$$\int \psi \, d\mu_{\varepsilon,p}(f) = \frac{1}{2\pi} \int v_{\varepsilon,p}(|f|) \Delta\psi \, dA. \tag{3.7}$$

For $p \ge 1$, the radial derivative is at most $1/\varepsilon$ on both pieces of (3.1); the radial function on $\mathbb{C}$ has the same global Lipschitz constant. For $0 < p < 1$, it is increasing, concave and zero at the origin. Consequently $|v(r) - v(s)| \le v(|r - s|)$ for $r, s \ge 0$. Moreover $v(h) \le h^p/(p\varepsilon^p)$: this is equality for $h \le \varepsilon$, while for $h \ge \varepsilon$ it follows from $\log x + 1/p \le x^p/p$ for $x \ge 1$. In either case $||f| - |g|| \le \eta$ and (3.7) give the claim. □

*Remark* 3.3 (Optimal scalar Lipschitz cost)*.* Among globally Lipschitz radial potentials whose radial profiles are $C^1$ for $r > 0$ and equal $\log r$ plus a constant for $r \ge \varepsilon$, any global Lipschitz constant is at least $1/\varepsilon$, because the derivative at $\varepsilon$ equals that number. Every potential (3.1) with $p \ge 1$ attains this minimum. For $p > 1$ these potentials are also $C^1$ on the plane; for $p = 1$ the origin has an allowed radial cusp. This is an optimality statement for the scalar potential within the specified class, not a lower bound on every algorithm for computing zero counts or on every integrated error. For the $q$-fold energy in Proposition 2.5, the natural exponent is $p = 2/q$: $q = 2$ is the borderline between this Lipschitz regime and the displayed Hölder regime for $q > 2$.

For the compact cubic disk cutoff, whose closed support disk lies inside $\Omega$, $\psi(z) = (1 - |z - C|^2/R^2)_+^3$, direct radial differentiation gives

$$\|\Delta\psi\|_1 = 12\pi \int_0^1 |-1 + 4s - 3s^2| \, ds = \frac{32\pi}{9}.$$

Thus (3.5) has allowance $16\eta/(9\varepsilon)$ for that cutoff. To turn a weighted integral into an unweighted count, one must also ensure that the cutoff is one on the entire low-value region; this requires a zero-free collar and is not part of the field-error hypothesis.

### 3.2 Quantitative localization near zeros

**Proposition 3.4** (Bias for a Lipschitz test)**.** *Assume (2.1). Let $a_1, \dots, a_k$ be all the zeros in $\Omega$, with multiplicities $m_j$. Choose disjoint closed disks $D_j \Subset \Omega$ centered at $a_j$ and positive constants $c_j$ such that*

$$|f(z)| \ge c_j |z - a_j|^{m_j} \quad (z \in D_j). \tag{3.8}$$

*Choose $\varepsilon$ smaller than the minimum of $|f|$ on the compact set $\overline{\Omega} \setminus \bigcup_j \operatorname{int} D_j$. Then for every Lipschitz function $\psi$ on $\overline{\Omega}$ and every $p > 0$,*

$$\left| \int \psi \, d\mu_{\varepsilon,p}(f) - \sum_{j=1}^{k} m_j \psi(a_j) \right| \le \operatorname{Lip}(\psi) \sum_{j=1}^{k} m_j (\varepsilon/c_j)^{1/m_j}. \tag{3.9}$$

*Proof.* The measure is supported in the union of the disks. Its mass in $D_j$ is $m_j$ by Proposition 3.1, and (3.8) restricts its support there to $|z-a_j| < (\varepsilon/c_j)^{1/m_j}$. Integrate $|\psi(z)-\psi(a_j)| \le \mathrm{Lip}(\psi)|z-a_j|$ on each disk. □

The constants in (3.8) exist by the nonvanishing analytic unit in the local factorization; their quantitative acquisition is additional information. The next result uses the smoothness already available for the tests in Theorem 3.2.

**Proposition 3.5** (Harmonic reproduction and second-order bias)**.** *Let $D = D(a, R)$, $R > 0$, and let $f$ be holomorphic near $\overline{D}$, with its only zero in $\overline{D}$ at $a$, of order $m \ge 1$. Put $b = \min_{\partial D} |f| > 0$. For $p > 0$, $0 < \varepsilon < b$, and every real harmonic $h$ on a neighborhood of $\overline{D}$,*

$$\int_D h \, d\mu_{\varepsilon,p}(f) = mh(a). \tag{3.10}$$

*If also $|f(z)| \ge c|z-a|^m$ on $\overline{D}$ for some $c > 0$, and $\psi$ is real $C^2$ near $\overline{D}$, then*

$$\left| \int_D \psi \, d\mu_{\varepsilon,p}(f) - m\psi(a) \right| \le \frac{m}{2} \|D^2\psi\|_{\infty,D} \frac{p}{p+2/m} (\varepsilon/c)^{2/m}. \tag{3.11}$$

*The Hessian norm is its real two-dimensional operator norm. No regular-value assumption on $\varepsilon$ is imposed.*

*Proof.* Choose a holomorphic $q$ near $\overline{D}$ with $h = \mathrm{Re}\, q$. The trace

$$T_q(w) = \frac{1}{2\pi i} \int_{\partial D} q(z) \frac{f'(z)}{f(z)-w} \, dz, \qquad |w| < b,$$

is holomorphic in $w$. By the residue theorem it is the sum of $q$ over all preimages of $w$ in $D$, with multiplicities, and $T_q(0) = mq(a)$. This does not label the preimages or assume them simple. The weighted area formula, applied to the real and imaginary parts, gives

$$\int_D q \, d\mu_{\varepsilon,p}(f) = \frac{p}{2\pi\varepsilon^p} \int_{|w|<\varepsilon} |w|^{p-2} T_q(w) \, dA_w = T_q(0).$$

The last equality is the angular mean-value identity followed by $p\varepsilon^{-p} \int_0^\varepsilon r^{p-1} dr = 1$. All integrals are absolutely convergent, since $q$ is bounded and the measure has mass $m$. Critical target values have area zero, including any on the cutoff circle; Proposition 3.1 excludes an additional interface measure. Taking real parts proves reproduction.

Every preimage of $w$ satisfies $|z-a| \le (|w|/c)^{1/m}$. The same area formula therefore yields

$$\int_D |z-a|^2 \, d\mu_{\varepsilon,p}(f) \le \frac{mp}{\varepsilon^p} \int_0^\varepsilon r^{p-1} (r/c)^{2/m} \, dr = m \frac{p}{p+2/m} (\varepsilon/c)^{2/m}. \tag{3.12}$$

Reproduction for real affine functions cancels the linear Taylor term of $\psi$ at $a$. The Taylor remainder on the convex disk is bounded by $\|D^2\psi\|_{\infty,D}|z-a|^2/2$. Integration proves (3.11). For $f(z) = c(z-a)^m$, $c > 0$, and $\psi(z) = |z-a|^2$, equality holds, including the displayed coefficient. □

Apply this proposition separately to the disks in Proposition 3.4; keep its complement floor and cutoff condition. The resulting bias is the sum of the right sides of (3.11). Combined with Theorem 3.2, it bounds the regularized approximate field against the exact divisor, even when the field error is known only on the support of the compact test. For fixed local data, $N > 0$, $M = \max_j m_j$, fixed $p \ge 1$, and $0 < \varepsilon \le 1$, the bound now has the form

$$A\eta/\varepsilon + B\varepsilon^{2/M}. \tag{3.13}$$

Taking $\varepsilon = \eta^{M/(M+2)}$ for sufficiently small $\eta$ gives $O(\eta^{2/(M+2)})$. For $0 < p < 1$, taking $\varepsilon = \eta^{pM/(pM+2)}$ gives $O(\eta^{2p/(pM+2)})$. If $N = 0$, the bias is zero for small $\varepsilon$. The earlier Lipschitz-only bias gives the valid weaker rates $1/(M+1)$ and $p/(pM+1)$. These optimized regularization bounds are upper estimates, not optimality statements about divisor perturbation.

### 3.3 Direct stability with complete cluster information

**Theorem 3.6** (Weighted divisor perturbation on a controlled disk)**.** *Let $R, c > 0$ and let $f, g$ be holomorphic near $\overline{D(a,R)}$. Suppose $f$ has exactly one zero there, at $a$ of order $m \geq 1$, and*

$$|f(z)| \geq c|z-a|^m \quad (z \in \overline{D}), \qquad b = \min_{\partial D} |f| > 0, \qquad \|f-g\|_{\infty,\overline{D}} \leq \eta < b.$$

*Then $g$ has exactly $m$ zeros $\zeta_1, \ldots, \zeta_m$ in $D$, counted with multiplicity. For every real $C^2$ function $\psi$ near $\overline{D}$,*

$$\left| \sum_{\nu=1}^{m} \psi(\zeta_\nu) - m\psi(a) \right| \leq |\nabla\psi(a)| R \log \frac{b}{b-\eta} + \frac{m}{2} \|D^2\psi\|_{\infty,D} (\eta/c)^{2/m}. \tag{3.14}$$

*No separation of the zeros of $g$, derivative-error estimate, or lower bound for its analytic unit is required.*

*Proof.* The case $\eta = 0$ is immediate. For $\eta > 0$, the classical Rouché theorem preserves the full count $m$, since $|g-f| < |f|$ on $\partial D$. At a zero of $g$, the interior error and the floor give $c|\zeta_\nu - a|^m \leq |f(\zeta_\nu)| \leq \eta$.

On the boundary, $g/f = 1 + u$ with $|u| \leq \eta/b < 1$. By compactness and holomorphy, $f$ is nonzero and $|u| < 1$ on a sufficiently thin collar of that circle. The convergent principal logarithm $\lambda = \log(1+u)$ is single valued and holomorphic on the collar, and its boundary values satisfy

$$|\lambda| \leq \sum_{k \geq 1} \frac{(\eta/b)^k}{k} = \log \frac{b}{b-\eta}.$$

Only this collar logarithm is used; there need be no logarithm of $g/f$ throughout $D$. The residue theorem and integration by parts around the closed circle give

$$\begin{aligned} \sum_{\nu=1}^{m} (\zeta_\nu - a) &= \frac{1}{2\pi i} \int_{\partial D} (z-a) \left( \frac{g'}{g} - \frac{f'}{f} \right) dz \\ &= -\frac{1}{2\pi i} \int_{\partial D} \lambda(z)\, dz. \end{aligned} \tag{3.15}$$

Consequently the modulus of this sum is at most $R \log[b/(b-\eta)]$. Summing the real Taylor formula at $a$ now bounds its linear term by this centroid allowance, and bounds the quadratic remainders by $m\|D^2\psi\|_{\infty,D}(\eta/c)^{2/m}/2$. This proves (3.14) with all multiplicities retained. □

For finitely many disjoint controlled disks $D_j = D(a_j, R_j)$, the same proof gives the sum of (3.14), with each $m_j, c_j, b_j, \eta_j$ retained. This compares the complete divisors in their union. To assert a statement about a larger domain, one must also exclude zeros of $g$ on its complement, for example by a field-error bound strictly below the positive floor of $|f|$ there.

For $0 \leq \eta \leq \theta b$, $\theta < 1$ fixed, $\log[b/(b-\eta)] \leq \eta/[(1-\theta)b]$. Thus the fixed-geometry rate in (3.14) is $O(\eta^{\min(1,2/m)})$ for tests with bounded $C^2$ norm. For $m \geq 2$, $f = z^m$, $g = z^m - \eta$ on the unit disk,

and $\psi = |z|^2$ give exactly $m\eta^{2/m}$. For $m = 1$, use $f = z$, $g = z - \eta$ and $\psi = \operatorname{Re} z$ to get exactly $\eta$. These fixed tests have smooth compact extensions on a larger disk, independent of $\eta$. Hence the exponent cannot be increased over this declared class. Translating the whole cluster to $d$, with $g = (z - d)^m$ and a linear test, also shows why a centroid term is necessary even though its Hessian vanishes. This is an exponent calibration, not an optimal constant or algorithmic lower bound.

The direct theorem has more information than Theorem 3.2: it controls the whole disk, the boundary and the analytic unit of $f$, and compares actual divisors. The latter theorem compares regularized measures using only the error on a test's support. Their rates are not claims under identical assumptions. Common multiplication of $f, g$ by $A \neq 0$ scales $c, b, \eta$ by $|A|$ and leaves (3.14) invariant. Constants are not asserted uniform as boundary floors vanish or independently chosen clusters merge.

The trace and area mechanisms are classical; the weighted area formula is stated in Simon [4, Chapter 2, §3]. Beauzamy's local polynomial bounds [5, Theorems 1 and 4] use a coefficient norm and give multiplicity-sensitive nearby-root estimates. Barlet [6, pp. 43–44] studies smooth functions summed over finite holomorphic fibers. Parusiński–Rainer [7, Theorem 1.1 and §11] study parameter-dependent polynomial roots in Sobolev and Wasserstein spaces. Those contracts differ from the disk-supremum and $C^2$-test estimate above. In particular, the monomial example has Wasserstein distance $\eta^{1/m}$ from its normalized zero measure to $\delta_0$, while its quadratic-test discrepancy is $\eta^{2/m}$. We claim an explicit specialization with proved constants, not priority over the trace or root-perturbation literature.

### 3.4 A smooth logarithmic alternative

Set

$$u_\varepsilon(f) = \tfrac{1}{2}\log(|f|^2 + \varepsilon^2), \qquad \widetilde{\mu}_\varepsilon(f) = \frac{1}{2\pi}\Delta u_\varepsilon(f) = \frac{\varepsilon^2|f'|^2}{\pi(|f|^2 + \varepsilon^2)^2}\, dA. \tag{3.16}$$

Unlike (3.2), this measure has no compact cutoff in target space. The chain rule proves (3.16) also at zeros, since its potential is smooth.

**Proposition 3.7** (Smooth-kernel boundary and field errors)**.** *Under (2.1), define* $V_\partial = (2\pi)^{-1}\int_{\partial\Omega} |d\arg f|$. *Then*

$$\widetilde{\mu}_\varepsilon(f)(\Omega) = \frac{1}{2\pi}\int_{\partial\Omega} \frac{|f|^2}{|f|^2 + \varepsilon^2}\, d\arg f, \tag{3.17}$$

$$|N - \widetilde{\mu}_\varepsilon(f)(\Omega)| \leq \frac{\varepsilon^2}{b^2 + \varepsilon^2} V_\partial, \qquad V_\partial \leq \frac{\operatorname{length}(\partial\Omega)\sup_{\partial\Omega}|f'|}{2\pi b}. \tag{3.18}$$

*For the fields and cutoff of Theorem 3.2,*

$$\left|\int \psi\, d\widetilde{\mu}_\varepsilon(f) - \int \psi\, d\widetilde{\mu}_\varepsilon(g)\right| \leq \frac{\eta}{4\pi\varepsilon}\|\Delta\psi\|_1. \tag{3.19}$$

*Also*

$$\widetilde{\mu}_\varepsilon(f)(\Omega) \leq \frac{9}{32\sqrt{3}\pi\varepsilon}\, W(f;\Omega). \tag{3.20}$$

*Proof.* On the boundary, the outward normal derivative of $u_\varepsilon$ is $|f|^2/(|f|^2 + \varepsilon^2)$ times that of $\log|f|$. The Cauchy–Riemann equations identify $\partial_n \log|f|\, ds = d\arg f$ with the specified orientations. Green's formula proves (3.17). Subtract the argument principle and take total variation to get

(3.18); pointwise, $|d\arg f| \le |f'|\,ds/b$. The scalar radial derivative $r/(r^2+\varepsilon^2)$ is bounded by $1/(2\varepsilon)$, giving (3.19) by integration by parts. Finally

$$\max_{r\ge 0} \frac{r}{(r^2+\varepsilon^2)^2} = \frac{9}{16\sqrt{3}\varepsilon^3},$$

attained at $r = \varepsilon/\sqrt{3}$. Comparing the density in (3.16) with $2|f'|^2/|f|$ proves (3.20). □

The boundary bias in (3.18) is an absolute error; no sign is asserted because $d\arg f$ need not have one sign along a general boundary. The improved factor two in (3.19) comes with that nonzero boundary bias. In contrast, the compact regularization has exact total mass below $b$.

### 3.5 Transport of the regularized count

Suppose $f(t,z)$ is smooth in real $t$ and holomorphic in $z$ on a neighborhood of $[t_0,T]\times\overline{\Omega}$, with no spatial-boundary zero. Its common boundary minimum $b_* > 0$ makes

$$\mu_{\varepsilon,p}(f(t))(\Omega) = N_{\mathbb{C}}(t_0) \quad (0<\varepsilon<b_*,\ t_0\le t\le T). \tag{3.21}$$

Indeed the argument-principle integer is continuous in $t$, and hence constant. For the smooth kernel one may also differentiate explicitly:

$$\partial_t \widetilde{\mu}_\varepsilon(f) = \frac{1}{2\pi}\Delta\left(\frac{\mathrm{Re}(\overline{f}f_t)}{|f|^2+\varepsilon^2}\right). \tag{3.22}$$

This identity is an equality of smooth densities. Under $f_t = -f_{zz}$ the numerator is $-\mathrm{Re}(\overline{f}f_{zz})$; integration gives its complete boundary flux, with no automatic sign. Interior collisions preserve the complex count in (3.21). The real-birth and discriminant balances in the next section measure different quantities.

## 4 Zero transport, real births, and complex event residues

This section distinguishes three quantities that agree in some double-zero models but are different in general: the number of complex zeros counted with multiplicity, the number of real zeros on a regular time slice, and the residue charge of a multiple-zero event. Every domain below is finite. In particular, no limiting contour, infinite product over zeros, or unproved assertion about zeros at arbitrarily large height is used. We use the backward-heat normalization

$$H_t = -H_{zz}. \tag{4.1}$$

When real variables are used, $H(t,x)$ is real and $H_x = H_z$ on the real slice. Joint real analyticity permits local complexification. All arguments involving complex time explicitly assume joint holomorphy on a neighborhood of the indicated closed product.

### 4.1 The local heat polynomial and its roots

**Lemma 4.1** (Elementary Hermite root calculation)**.** *For $m \ge 1$ define*

$$p_m(s,w) = \sum_{k=0}^{\lfloor m/2\rfloor} \frac{m!(-s)^k w^{m-2k}}{k!(m-2k)!}.$$

*Then $p_m(1,\cdot)$ has $m$ distinct real roots. The polynomial $p_m(-1,\cdot)$ has no real root when $m$ is even and exactly one real root, a simple root at zero, when $m$ is odd. If $\lambda_1,\dots,\lambda_m$ are the roots of the monic probabilists' Hermite polynomial, then the roots of $p_m(1,\cdot)$ are $\sqrt{2}\lambda_1,\dots,\sqrt{2}\lambda_m$.*

*Proof.* Define

$$\mathrm{He}_m(u) = (-1)^m e^{u^2/2}\frac{d^m}{du^m}e^{-u^2/2}.$$

Differentiating $e^{u\xi-\xi^2/2}$ in $\xi$, or expanding its two exponentials, gives

$$\mathrm{He}_m(u) = \sum_{k=0}^{\lfloor m/2\rfloor}\frac{m!(-1)^k u^{m-2k}}{2^k k!(m-2k)!}, \qquad p_m(1,w) = 2^{m/2}\,\mathrm{He}_m(w/\sqrt{2}).$$

Repeated integration by parts, whose boundary terms vanish by Gaussian decay, shows that

$$\int_{\mathbb{R}} \mathrm{He}_m(u)q(u)e^{-u^2/2}\,du = 0 \quad (\deg q < m).$$

Suppose that $\mathrm{He}_m$ has fewer than $m$ distinct real zeros of odd multiplicity. Let $q$ be the product of the linear factors at those zeros, once each. Then $\deg q < m$, while $\mathrm{He}_m\, q$ has constant sign on the real line apart from its zeros and is not identically zero. Its Gaussian integral cannot vanish, a contradiction. Degree $m$ forces all its roots to be real and simple. Finally $p_m(-1,w)$ is a polynomial in $w^2$ with strictly positive coefficients if $m$ is even; if $m$ is odd it is $w$ times such a polynomial, with a strictly positive constant term. This proves the negative-time assertion as well. □

**Theorem 4.2** (Local analytic splitting and isolation)**.** *Suppose $H$ is holomorphic near $(t_*,z_*)$, satisfies (4.1), and*

$$H(t_*, z_*+w) = c_m w^m + O(w^{m+1}), \qquad c_m \neq 0.$$

*There is a proper local cluster of exactly $m$ spatial zeros, counted with multiplicity. With $s = t-t_* \neq 0$ sufficiently small, all its roots are simple. More precisely, after $s=\varepsilon^2$, they can be written*

$$z_j(t_*+\varepsilon^2) - z_* = \varepsilon(\sqrt{2}\lambda_j + O(\varepsilon)), \qquad 1\le j\le m, \tag{4.2}$$

*with the parenthesized functions holomorphic in $\varepsilon$. For $m\ge 2$, $(t_*,z_*)$ is consequently an isolated common zero of $(H,H_z)$ in complex two-space.*

*If $t_*, z_*$ are real and $H$ is of real type, all $m$ roots are real for small positive $s$. For small negative $s$, exactly $m \bmod 2$ of them are real. Thus the local number of real zeros increases by $2\lfloor m/2\rfloor$ as real time passes through the event.*

*Proof.* Write $H(t_*, z_*+w) = \sum_{n\ge m} c_n w^n$. The heat equation identifies the convergent bivariate Taylor coefficients as

$$H(t_*+s, z_*+w) = \sum_{k,l\ge 0}\frac{(-1)^l(k+2l)!}{k!l!}c_{k+2l}s^l w^k.$$

Every nonzero term has weighted degree at least $m$, when $s$ has weight two and $w$ weight one. Substitution $s=\varepsilon^2$, $w=\varepsilon y$ therefore gives a holomorphic extension to $\varepsilon=0$ of

$$F(\varepsilon,y) = \varepsilon^{-m}H(t_*+\varepsilon^2, z_*+\varepsilon y), \qquad F(0,y) = c_m p_m(1,y).$$

Convergence is uniform on every sufficiently small $\varepsilon$-disk and every fixed bounded $y$-disk. Lemma 4.1 and the holomorphic implicit function theorem give $m$ distinct root branches $y_j(\varepsilon) = \sqrt{2}\lambda_j + O(\varepsilon)$.

Choose a fixed small spatial circle around $z_*$ on which $H(t_*, \cdot)$ is nonzero and whose interior contains just its order-$m$ zero. Uniform continuity on that circle, followed by Rouché's theorem, preserves the count $m$ for all sufficiently small complex $s$. The displayed branches lie inside this circle and account for every root. Since $F_y$ remains nonzero at them, $H_z$ is nonzero whenever $\varepsilon \neq 0$. This proves isolation in complex two-space.

For real positive $\varepsilon$, each branch is real by uniqueness and complex conjugation, so the positive-time conclusion follows. At negative real $s$, take $\varepsilon = i\sqrt{|s|}$. If $\lambda_j \neq 0$, (4.2) has a nonzero imaginary leading term of order $\sqrt{|s|}$, and its error is $O(|s|)$, so that root is not real. There is no zero Hermite root when $m$ is even. When $m$ is odd there is exactly one, because parity gives a zero and all Hermite roots are simple. For its branch, $w_0(\varepsilon) = \varepsilon y_0(\varepsilon) = O(\varepsilon^2)$. Both $w_0(\varepsilon)$ and $w_0(-\varepsilon)$ are roots of the same function $H(t_* + \varepsilon^2, z_* + w)$ and correspond to the unique $y$-root near zero: the second has normalized coordinate $w_0(-\varepsilon)/\varepsilon = -y_0(-\varepsilon) = O(\varepsilon)$. Hence they coincide. The function $w_0$ is even in $\varepsilon$ and is holomorphic in $s$. Reality on positive real $s$ makes its Taylor coefficients real, so it remains real for negative real $s$. It is the sole real root there. □

## 4.2 An angular-current formula with every wall term

Give the real $(t, x)$-plane the orientation $dt \wedge dx$. Wherever $J = (H, H_x) \neq 0$, put

$$\alpha = \frac{H\, dH_x - H_x\, dH}{H^2 + H_x^2}, \qquad \beta = -\frac{\alpha}{2\pi}. \tag{4.3}$$

Locally $\alpha = d\arg(H + iH_x)$, so $d\alpha = d\beta = 0$ away from joint zeros. The sign in $\beta$ is chosen to assign a positive real charge to a backward-heat event. On a spatial cap its coefficient is

$$\beta\big|_{t=\text{constant}} = \frac{H_x^2 - HH_{xx}}{2\pi(H^2 + H_x^2)}\, dx. \tag{4.4}$$

**Lemma 4.3** (Phase increments and signed crossings)**.** *Let an oriented analytic path carry a nonzero real vector $(A, B)$, with $A$ nonzero at its endpoints and not identically zero. At an isolated zero $v_j$ of $A$, put*

$$q_j = \operatorname{sgn} B(v_j)\, \frac{\operatorname{sgn} A(v_j + \delta) - \operatorname{sgn} A(v_j - \delta)}{2},$$

*where $\delta > 0$ is small enough to isolate that zero. Then*

$$\int \frac{A\, dB - B\, dA}{A^2 + B^2} = [\arctan(B/A)]_{\text{start}}^{\text{end}} - \pi \sum_j q_j. \tag{4.5}$$

*Even-order tangencies contribute zero. At a transverse crossing, $q_j = \operatorname{sgn}(BA')(v_j)$.*

*Proof.* Off the zeros of $A$, the differential of the principal arctangent $\arctan(B/A) \in (-\pi/2, \pi/2)$ is the integrand. At a zero of $A$, $B$ is nonzero. The one-sided limits of the arctangent have difference $\pi q_j$: they are equal at an even tangency and differ by the stated signed multiple of $\pi$ at an odd crossing. Summing the integrals on the complementary path intervals gives (4.5). The formula is also the increment of a continuous argument of $A + iB$. □

**Proposition 4.4** (Cap count and wall correction)**.** *Let*

$$\mathcal{R} = \{(t, x) : a < t < b, \quad \ell(t) < x < r(t)\}$$

*be a bounded moving rectangle, with analytic walls $\ell < r$ on a neighborhood of $[a, b]$. Assume $H$ is real analytic near its closure, $J = (H, H_x)$ is nonzero on its whole boundary, and $H$ is nonzero*

*at the four corners. Let $N_a, N_b$ count the zeros of $H$ on the two open spatial caps; these zeros are simple. For either wall $c = \ell, r$, set $F_c(t) = H(t, c(t))$ and*

$$Q_c = \sum_{F_c(t_j)=0} \operatorname{sgn} H_x(t_j, c(t_j)) \, \frac{\operatorname{sgn} F_c(t_j + \delta_j) - \operatorname{sgn} F_c(t_j - \delta_j)}{2}. \tag{4.6}$$

*The sums are finite, and*

$$\frac{1}{2\pi} \oint_{\partial \mathcal{R}} \alpha = \frac{N_a - N_b - Q_\ell + Q_r}{2}, \qquad \oint_{\partial \mathcal{R}} \beta = \frac{N_b - N_a + Q_\ell - Q_r}{2}. \tag{4.7}$$

*The orientation traverses the lower wall with increasing $t$, the late cap with increasing $x$, the upper wall with decreasing $t$, and the early cap with decreasing $x$.*

*Proof.* Analyticity and nonzero corner values make each wall function nonzero identically and give finitely many wall zeros on the compact interval. At such a zero, boundary regularity gives $H_x \neq 0$. The same argument on each cap gives finitely many simple spatial zeros. Apply Lemma 4.3 to each of the four edges with $(A, B) = (H, H_x)$. Along a forward cap, the signed crossing number is $+1$ at every root, since $A' = H_x = B$. Along a forward wall it is the corresponding summand of $Q_c$. The four endpoint arctangent differences telescope. The remaining contributions, with the declared orientations, are $-\pi Q_\ell - \pi N_b + \pi Q_r + \pi N_a$. This proves both identities. □

*Remark* 4.5 (A single cap still counts ordinary zeros). For fixed walls $\ell < r$ with $H(t, \ell) H(t, r) \neq 0$, let

$$B(t) = \int_\ell^r \beta, \qquad \theta_c(t) = \arctan \frac{H_x(t, c)}{H(t, c)}.$$

On a regular cap, Lemma 4.3 gives exactly

$$B(t) = \frac{N_t}{2} + \frac{\theta_\ell(t) - \theta_r(t)}{2\pi}. \tag{4.8}$$

Thus simplicity of every cap root does not imply that the cap integral vanishes. If $H$ never vanishes on either wall, the wall contribution to $\oint \beta$ is $[\theta_r(b) - \theta_r(a) - \theta_\ell(b) + \theta_\ell(a)]/(2\pi)$; it cancels precisely the endpoint terms of the two cap formulas.

**Theorem 4.6** (Complete real birth balance)**.** *Assume the hypotheses of Proposition 4.4, the heat equation (4.1), and no identically zero spatial slice in the analytic neighborhood. The interior event set $\{H = H_x = 0\}$ is finite. If $m(T, x_*)$ is the exact spatial multiplicity at an event, then*

$$\operatorname{ind}_{(T, x_*)}(H, H_x) = -\left\lfloor \frac{m(T, x_*)}{2} \right\rfloor, \tag{4.9}$$

$$\oint_{\partial \mathcal{R}} \beta = \sum_{(T, x_*) \in \mathcal{R}:\, H = H_x = 0} \left\lfloor \frac{m(T, x_*)}{2} \right\rfloor, \tag{4.10}$$

$$N_b - N_a + Q_\ell - Q_r = 2 \sum_{(T, x_*) \in \mathcal{R}:\, H = H_x = 0} \left\lfloor \frac{m(T, x_*)}{2} \right\rfloor. \tag{4.11}$$

*Here the local index is the counterclockwise winding of $(H + iH_x)/|H + iH_x|$ on a small positively oriented loop.*

*Proof.* Each event has finite spatial multiplicity by the nonzero-slice hypothesis and is isolated by Theorem 4.2. Boundary regularity, compactness, and analyticity then make the event set finite: an infinite set would have an accumulation point in the compact closure, which could be neither a regular point nor an isolated event.

For one event, choose a small fixed real spatial interval enclosing its local cluster and a small time interval symmetric about $T$. Its walls have $H \neq 0$, its time caps are regular, and its sole event is the given one. Theorem 4.2 gives cap counts $m \bmod 2$ and $m$. Proposition 4.4 therefore gives local winding $((m \bmod 2) - m)/2 = -\lfloor m/2 \rfloor$, proving (4.9). Remove disjoint small neighborhoods of the finitely many events from $\mathcal{R}$. On the remainder $d\beta = 0$. Stokes' theorem, with the inner boundaries oppositely oriented, gives (4.10). Combining it with (4.7) proves (4.11). □

For a transverse wall crossing the formula can also be checked by ordinary root motion. A simple root satisfies $x'(t) = H_{xx}/H_x$, so

$$Q_c = \sum_{H(t,c(t))=0} \operatorname{sgn}\{H_x(-H_{xx} + c'H_x)\}.$$

A positive lower-wall crossing removes a root from the interval; a positive upper-wall crossing adds one. This is the sign convention in (4.11). The sign-change definition (4.6), unlike this transverse formula, also handles higher-order contacts.

### 4.3 Conservation of the ordinary complex divisor

**Proposition 4.7** (Multiplicity conservation on a proper spatial domain)**.** *Let $\Omega$ be a bounded domain with piecewise smooth boundary. Suppose $H(t,\cdot)$ is holomorphic on a neighborhood of $\overline{\Omega}$, continuously with its first spatial derivative in $t \in [a,b]$, and $H(t,z) \neq 0$ for $z \in \partial\Omega$. Then*

$$N_{\mathbb{C}}(t) = \frac{1}{2\pi i} \oint_{\partial\Omega} \frac{H_z(t,z)}{H(t,z)}\, dz \tag{4.12}$$

*is independent of $t$, with every root counted with its multiplicity. No heat equation is needed.*

*Proof.* The argument principle makes (4.12) an integer. The nonzero boundary condition and compactness give a positive common boundary margin. Hence the displayed integral is continuous in $t$, and a continuous integer-valued function on an interval is constant. □

In particular, collision and splitting are not creation or destruction of the complex divisor. The $m$ descendants in Theorem 4.2 have the same total multiplicity as the initial order-$m$ zero. Their realness can change, which is why (4.11) carries an event term while (4.12) does not. A domain that permits zeros to cross its spatial boundary must retain the signed boundary crossings; the properness hypothesis is exactly what eliminates them here.

### 4.4 Preparation and the discriminant residue

Let $D \subset \mathbb{C}$ be a bounded disk or rectangle and $\Omega_R = \{z : |z - z_c| < R\}$. Suppose $H$ is holomorphic on a neighborhood of $\overline{D} \times \overline{\Omega}_R$ and

$$H(t,z) \neq 0 \qquad (t \in \overline{D},\ |z - z_c| = R). \tag{4.13}$$

The number $n$ of roots in $\Omega_R$, including multiplicity, is then constant. We call this a proper finite spatial cluster.

**Lemma 4.8** (Preparation from contour moments)**.** *Under (4.13), there is a unique monic polynomial $P(t,z)$ of degree $n$ in $z$, with coefficients holomorphic in $t$, whose roots are precisely the selected cluster. Moreover $H(t,z) = U(t,z)P(t,z)$, where $U$ is holomorphic and nonvanishing on a neighborhood of the closed product.*

*Proof.* For $1 \le k \le n$, set

$$S_k(t) = \frac{1}{2\pi i} \oint_{|\zeta - z_c| = R} \zeta^k \frac{H_z(t,\zeta)}{H(t,\zeta)}\, d\zeta.$$

Residues identify these holomorphic functions as the power sums of the cluster roots. Newton's identities recursively express the elementary symmetric functions as polynomials in $S_1, \dots, S_n$; these are the coefficients of the required monic $P$. At each fixed $t$, the quotient $H/P$ has only removable singularities and is nonzero throughout the disk. Its joint holomorphy follows from

$$U(t,z) = \frac{1}{2\pi i} \oint_{|\zeta - z_c| = R} \frac{H(t,\zeta)}{P(t,\zeta)(\zeta - z)}\, d\zeta.$$

For each $t$, this is precisely the removable quotient by Cauchy's formula. On a collar of the spatial boundary it agrees with the ordinary quotient, giving the extension there. Strict nonvanishing on the compact wall persists after a small enlargement, which also gives the asserted neighborhood. Uniqueness follows from the prescribed root multiset and monicity. □

Define the finite discriminant by

$$\Delta_P(t) = \prod_{i<j} (z_i(t) - z_j(t))^2, \qquad \Delta_P = 1 \text{ if } n \le 1. \tag{4.14}$$

It is a symmetric polynomial in the roots and therefore holomorphic in the coefficients of $P$, even through collisions and without a global choice of root labels.

**Proposition 4.9** (The local heat residue)**.** *For the local cluster at an order-$m$ heat event from Theorem 4.2, put $q_m = \binom{m}{2}$. Its discriminant has*

$$\Delta(t_* + s) = d_m s^{q_m} + O(s^{q_m+1}), \qquad d_m = 2^{q_m} \prod_{i<j} (\lambda_i - \lambda_j)^2 > 0. \tag{4.15}$$

*Consequently $d \log \Delta$ has residue $q_m$ at the event time.*

*Proof.* Substitute the $m$ root branches (4.2) into (4.14). Each squared difference has leading term $2(\lambda_i - \lambda_j)^2 \varepsilon^2$. Their product is $d_m \varepsilon^{2q_m}(1 + O(\varepsilon))$. The discriminant is holomorphic in $s = \varepsilon^2$, so its Taylor series in $\varepsilon$ is even. This improves the remainder to $O(\varepsilon^2)$ and proves (4.15). Taking the logarithmic derivative gives the residue. The coefficient is positive because the Hermite roots are real and distinct; the original scalar factor $c_m$ does not enter a monic-root discriminant. □

**Theorem 4.10** (Finite complex event charge)**.** *Assume (4.13), the heat equation, and*

$$(H, H_z) \neq (0,0) \qquad \text{on } \partial D \times \overline{\Omega}_R.$$

*Then there are finitely many events in $D \times \Omega_R$, and*

$$Q_{\mathbb{C}} := \frac{1}{2\pi i} \oint_{\partial D} d \log \Delta_P = \sum_{(t,z) \in D \times \Omega_R :\, H = H_z = 0} \binom{m(t,z)}{2}. \tag{4.16}$$

*The boundary of $D$ has its positive complex orientation. In particular, all interior complex event residues have the same positive sign.*

*Proof.* The hypotheses make $\Delta_P$ nonzero on $\partial D$. Thus it is not identically zero and has finitely many zeros in the closed time domain. At a zero time, the spatial cluster is finite. Small disjoint spatial disks about its distinct roots give separate monic cluster polynomials. The discriminant of their product is the product of their discriminants times the squared pairwise resultants. The latter are nonzero at the chosen time because roots in distinct disks do not coincide. Hence the order of the full discriminant is the sum of the local orders. Proposition 4.9 identifies each order as $\binom{m}{2}$. The scalar argument principle now proves (4.16). □

**Proposition 4.11** (Agreement with the complex boundary degree)**.** *Under the hypotheses of Theorem 4.10, give $D \times \Omega_R$ the orientation $d\operatorname{Re} t \wedge d\operatorname{Im} t \wedge d\operatorname{Re} z \wedge d\operatorname{Im} z$. Then*

$$\deg\left(\frac{(H, H_z)}{\sqrt{|H|^2 + |H_z|^2}}\right)\Big|_{\partial(D\times\Omega_R)} = \frac{1}{2\pi i}\oint_{\partial D} d\log \Delta_P. \tag{4.17}$$

*Proof.* If $n = 0$, then $H$ has no zero on the whole product, the normalized jet extends across it, and both sides are zero. Assume henceforth $n \geq 1$. Here we use the elementary degree facts that a boundary map into the unit sphere has homotopy-invariant integer degree, and that its degree is the signed number of inverse images of a nearby regular value. The latter follows by excising neighborhoods of the inverse images and applying Stokes' theorem to the normalized sphere volume form; each invertible derivative contributes the sign of its real determinant. These facts apply to the piecewise smooth product boundary by rounding corners in its nonvanishing collar.

By preparation,

$$\begin{pmatrix} H \\ H_z \end{pmatrix} = \begin{pmatrix} U & 0 \\ U_z & U \end{pmatrix}\begin{pmatrix} P \\ P_z \end{pmatrix}.$$

Since the product of two disks or a rectangle and a disk is simply connected and $U$ is nonzero, $U$ has a holomorphic logarithm there. The invertible matrices

$$M_v = \begin{pmatrix} e^{v\log U} & 0 \\ vU_z & e^{v\log U} \end{pmatrix}, \qquad 0 \leq v \leq 1,$$

give a nonzero boundary homotopy from $(P, P_z)$ to $(H, H_z)$. It suffices to compute the degree of $F = (P, P_z)$.

Choose a regular target value $(\eta, \xi)$ with $\xi \neq 0$ and norm smaller than $\min_{\partial(D\times\Omega_R)} |F|$, and small enough for the Rouché comparison below. Such choices exist by the regular-value theorem, and they have no preimage on the product boundary. Uniform Rouché comparison on the spatial wall keeps every root of $P(t, \cdot) - \eta$ inside $\Omega_R$ for all $t \in \overline{D}$. Consider the scalar resultant

$$R_{\eta,\xi}(t) = \operatorname{Res}_z(P(t, z) - \eta, P_z(t, z) - \xi).$$

Its zeros are exactly the time coordinates of the inverse images of $(\eta, \xi)$. At such an inverse image $P_z = \xi \neq 0$, so $P(t, z) - \eta = 0$ has a local branch $z = r(t)$. The corresponding factor $g(t) = P_z(t, r(t)) - \xi$ has

$$g'(t) = P_{zt} - P_{zz}P_t/\xi = -\det(D_{(t,z)}F)/\xi \neq 0.$$

Thus this solution contributes one to the zero order of the resultant; several solutions at the same time contribute additively. All other local factors are nonzero. It follows that the argument-principle count for $R_{\eta,\xi}$ equals the number of regular inverse images. Every one contributes $+1$ to degree because the real Jacobian of a complex-linear invertible derivative is $|\det_{\mathbb{C}} DF|^2 > 0$.

At target zero the resultant is

$$R_{0,0}(t) = \prod_i P_z(t, z_i) = (-1)^{n(n-1)/2} \Delta_P(t).$$

The formula holds first for distinct roots and then everywhere by holomorphy. Small target perturbations keep the resultant nonzero on $\partial D$, so its winding is unchanged. The number of inverse images, and hence the degree, equals the winding of $\Delta_P$, as asserted. □

*Remark* 4.12 (Dimension and multiplicity distinctions). The integral (4.12) counts ordinary zeros on a spatial complex curve. The real angular current (4.3) is a one-form on a two-dimensional real spacetime and counts births with weight $\lfloor m/2 \rfloor$. Formula (4.17) is a degree on a three-dimensional boundary in complex two-space and has weight $\binom{m}{2}$. Its reduction to a one-dimensional time contour is through the discriminant, not through the ordinary divisor count. The two event weights agree for $m = 2$ but already differ for $m = 3$.

## 4.5 Cap decompositions and endpoint terms

**Proposition 4.13** (The complete scalar cap balance). *Let $D = (a, b)+i(-Y, Y)$ satisfy Theorem 4.10. Define forward vertical cap contributions by*

$$C_t = \frac{1}{2\pi} \int_{-Y}^{Y} \frac{d}{ds} \arg \Delta_P(t + is)\, ds, \qquad t = a, b,$$

*using any continuous argument along the indicated nonzero segment. Let $S$ be the sum of the argument increments, divided by $2\pi$, on the lower horizontal edge traversed from $a - iY$ to $b - iY$ and the upper horizontal edge traversed from $b + iY$ to $a + iY$. Then*

$$Q_{\mathbb{C}} = C_b - C_a + S. \tag{4.18}$$

*Subtracting $d \log \Delta_0$ for a holomorphic nonvanishing reference $\Delta_0$ on a neighborhood of $\overline{D}$ changes the individual edge contributions but leaves their total unchanged.*

*Proof.* The argument increment around a nonzero closed loop is the imaginary part of its logarithmic-derivative integral. The real part is the increment of $\log |\Delta_P|$ and telescopes to zero. Splitting the positively oriented boundary into its four edges gives the stated signs. A nonvanishing holomorphic reference on the simply connected rectangle has a holomorphic logarithm, whose closed-contour integral is zero. □

Consequently, for a family of complete domains carrying the same interior charge, genuinely proved limits $C_b \to 0$ and $S \to 0$ would imply $C_a \to -Q_{\mathbb{C}}$. This implication is a transport identity, not a proof of either limit. A late cap with only simple roots merely has $\Delta_P \neq 0$; its argument may still vary. A reference defined only locally, or one with its own zeros or poles, instead brings its transition or divisor terms and is not covered by the final assertion of Proposition 4.13.

**Proposition 4.14** (An explicitly retained endpoint indentation). *Suppose a scalar discriminant has an event at a boundary time $t_0$, with $\Delta(t) = (t - t_0)^q g(t)$, $q \geq 1$, and $g$ holomorphic and nonzero near $t_0$. On an oriented circular arc $t = t_0 + \varepsilon e^{i\theta}$, with $\theta$ running from $\theta_0$ to $\theta_1$ in the declared orientation,*

$$\frac{1}{2\pi i} \int_{\text{arc}} d \log \Delta = \frac{q(\theta_1 - \theta_0)}{2\pi} + O(\varepsilon). \tag{4.19}$$

*A clockwise semicircular indentation therefore contributes $-q/2 + O(\varepsilon)$. The argument principle applies to the indented domain with this arc included, counting only the events left in its interior.*

*Proof.* Write $d\log\Delta = q\,dt/(t-t_0) + (g'/g)dt$. The first integral is $iq(\theta_1-\theta_0)$; the second is $O(\varepsilon)$ because $g'/g$ is bounded and the arc has length $O(\varepsilon)$. Division by $2\pi i$ gives (4.19). The final assertion is the ordinary argument principle on the boundary-regular indented domain. □

*Remark* 4.15 (No silent endpoint convention). A real multiple zero on the initial cap violates the boundary-jet hypothesis of Theorem 4.6. One may start at an earlier regular time, or excise a prescribed small neighborhood of the endpoint and retain the resulting oriented arc in the real angular integral. Stokes' theorem remains exact on this excised domain. Its real angular arc is not automatically given by the scalar discriminant formula (4.19); these are different currents on different spaces. In particular, no universal "half charge" for an unspecified endpoint regularization is assumed here.

An order-$m$ real zero on the initial cap, counted with its spatial multiplicity, already contributes $m$. Its $m$ simple real descendants at positive time also contribute $m$. Thus ordinary multiplicity counts at these two times do not detect the initial multiplicity. Moreover (4.8) cannot be evaluated at that endpoint by replacing its regular-root count with a multiplicity count: the angular current itself is undefined at the joint zero. This is the exact reason that an all-zero cap difference alone is not an endpoint simplicity theorem.

### 4.6 Exact controls

For $H(t,z) = z^2 - 2t$, a sufficiently wide proper spatial window has two complex zeros on either regular time cap, whereas its real counts are zero before $t=0$ and two after it. Its real charge is one and $\Delta = 8t$ gives complex charge one. For $H(t,z) = z^3 - 6tz$, the real counts are one and three, so its real charge is again one, while $\Delta = 864t^3$ gives complex charge three. Both controls satisfy the analytic and heat hypotheses exactly. They verify the normalization and illustrate that none of the conservation identities excludes an event. No numerical approximation is used in any proof of this section.

*Remark* 4.16 (Scope). These identities hold for every heat family satisfying the stated finite-domain hypotheses. The explicit polynomial controls show that they are compatible with multiple-zero events. An exclusion theorem requires an additional analytic estimate on the complete boundary integral, including every cap, wall, and endpoint term, that contradicts the positive event sum. Neither simplicity of a late cap nor conservation of its ordinary complex multiplicity supplies that estimate.

## 5 Cover topology and the choice of differential

The energy identity and the topology of a normalized cover are compatible, but they are not interchangeable. This distinction is visible without any asymptotic limit.

**Proposition 5.1** (Euler characteristic of a normalized root cover)**.** *Let $\Omega$ be a disk and suppose $f$ is holomorphic near $\overline{\Omega}$, nonzero on its boundary, with distinct interior zeros $a_1,\dots,a_k$ of orders $m_1,\dots,m_k$. On the normalization $X_q$ of $w^q = f(z)$,*

$$\chi(X_q) = q - \sum_{j=1}^{k}(q - \gcd(q,m_j)). \tag{5.1}$$

*For $q=2$, writing $B_{\mathrm{odd}} = \#\{j : m_j \text{ is odd}\}$ and $N = \sum_j m_j$, this becomes*

$$\chi(X_2) = 2 - B_{\mathrm{odd}}, \qquad \chi(X_2) - (2-N) = 2\sum_j \lfloor m_j/2 \rfloor. \tag{5.2}$$

*Proof.* The local normalization described in Proposition 2.5 has $d_j = \gcd(q, m_j)$ points above $a_j$, each with ramification index $q/d_j$. Triangulate the closed disk with the zeros as vertices and with no other branch values on edges. Each open edge and face has $q$ lifts; ordinary vertices have $q$ lifts, while $a_j$ has $d_j$. Thus the lifted Euler characteristic is $q\chi(\Omega) - \sum_j (q - d_j)$, proving (5.1). For $q = 2$, $d_j$ is one for odd $m_j$ and two for even $m_j$. Subtracting $2 - N$ gives $N - B_{\text{odd}} = 2\sum_j \lfloor m_j/2 \rfloor$. □

The proof includes disconnected covers, for example the two components of the normalized curve $w^2 = z^2$. In a proper analytic family with fixed complex zero count $N$, any regular fiber with $N$ simple zeros has $\chi(X_2) = 2 - N$. Before and after a collision the regular complex fibers therefore have the same Euler characteristic. The excess in (5.2) occurs at the *normalized singular fiber*, not as a permanent difference between the two regular fibers. The unnormalized singular curve is a different topological space.

**Proposition 5.2** (Reciprocal differential at a zero)**.** *On the normalized double cover $w^2 = f(z)$, the differential $\omega = dz/w$ is regular above a simple zero. At a zero of order $m = 2r$ it has a pole of order $r$ on each of the two local components. At a zero of order $m = 2r + 1$ it has a pole of order $2r$ when $r \geq 1$. Its reciprocal energy, denoted $\mathcal{E}(f; \Omega)$, is*

$$\frac{i}{2} \int_{X_2} \omega \wedge \overline{\omega} = 2 \int_{\Omega} \frac{dA}{|f|}, \tag{5.3}$$

*where both sides are interpreted locally or as extended integrals. This energy is finite at a simple zero, logarithmically divergent at a double, and power-divergent at higher multiplicity.*

*Proof.* Write $f(z) = z^m g(z)$ after translating the zero, with $g(0) \neq 0$ and with a local holomorphic square root of $g$. For $m = 2r$ the normalized charts have $w = \pm z^r \sqrt{g}$, so $\omega = \pm z^{-r} g^{-1/2}\, dz$. For $m = 2r + 1$, use $z = \xi^2$ and $w = \xi^{2r+1}\sqrt{g(\xi^2)}$ to obtain

$$\omega = 2\xi^{-2r} g(\xi^2)^{-1/2}\, d\xi.$$

Summing the two sheetwise densities gives (5.3). The radial comparison $|f(z)| \asymp |z|^m$ reduces integrability to $\int_0^\delta r^{1-m}\, dr$. □

By contrast, the derivative differential $\eta_2 = 2\, dw$ is regular at all multiplicities. Its finite energy counts zeros only after the localization and normalization in Theorem 2.2. At a pure double $f = Az^2$, the residues of $dz/w$ are $\pm A^{-1/2}$; they are not integer residues. The integer residue of $d \log f$ counts ordinary multiplicity, while that of a cluster discriminant counts collision multiplicity. These three uses of "residue" have different normalizations and different consumers.

**Proposition 5.3** (Amplitude and analytic-unit dependence)**.** *Multiplication of a heat field $H$ by a nonzero constant $c$ preserves its zeros, their trajectories and all pair distances. Nevertheless*

$$\omega_{cH} = c^{-1/2}\omega_H, \qquad \oint \omega_{cH} = c^{-1/2} \oint \omega_H,$$
$$\mathcal{E}(cH; \Omega) = |c|^{-1}\mathcal{E}(H; \Omega), \qquad W(cH; \Omega) = |c| W(H; \Omega),$$

*with a consistent square-root choice and cycles identified through the natural cover isomorphism. More generally, if $H = UP$ on a simply connected disk, with $U$ nonvanishing, the identification $w = \sqrt{U}\, v$ sends the cover to $v^2 = P$, but sends the physical differential to $dz/(\sqrt{U}\, v)$.*

*Proof.* The asserted transformations follow by substitution in the differentials and energy densities. A holomorphic nonvanishing function on a simply connected disk has a holomorphic logarithm, hence a coherent square root. □

Thus root positions determine the finite branched covering data, but not the unnormalized physical periods or energies without the analytic unit. The normalized period matrix of a separately specified compact Riemann surface is another object. The argument above concerns the physical differential and the bordered cover, whose relative periods and boundary terms remain part of its energy. For the canonical theta field the amplitude is fixed by its source; the proposition is an information test, not permission to change that normalization.

# 6 Finite pair matrices and the complete heat dynamics

A pair statistic becomes useful for this problem only after its population, spatial normalization, singular weights, and exterior contribution have been specified. This section works with a fixed finite population. Its matrix is the inverse-square graph Laplacian, which is also the Hessian of the logarithmic pair energy. The subsequent canonical theta application proves convergence to a clock matrix by a direct complete-source concentration estimate. No random ensemble or statistical universality assumption enters either argument.

## 6.1 A finite inverse-square matrix

For an ordered configuration $x = (x_1, \dots, x_m)$, where $x_1 < \cdots < x_m$ and $m \geq 2$, set

$$S_i(x) = \sum_{j \neq i} \frac{1}{x_i - x_j}, \qquad W_{ij}(x) = \frac{1}{(x_i - x_j)^2} \quad (i \neq j). \tag{6.1}$$

Define the real symmetric matrix $\mathcal{L}(x)$ by

$$\mathcal{L}(x)_{ij} = -W_{ij}(x) \quad (i \neq j), \qquad \mathcal{L}(x)_{ii} = \sum_{j \neq i} W_{ij}(x). \tag{6.2}$$

For every $v \in \mathbb{R}^m$,

$$v^{\mathsf{T}} \mathcal{L}(x) v = \sum_{i<j} \frac{(v_i - v_j)^2}{(x_i - x_j)^2}. \tag{6.3}$$

In particular $\mathcal{L}(x)$ is positive semidefinite, its kernel is exactly the constant vectors, and all its other eigenvalues are positive. The positivity is a statement about a configuration at one time; by itself it gives no sign for the time derivative of this matrix.

**Theorem 6.1** (Finite clock-matrix comparison)**.** *Let $c_i = c_0 + i - 1$, $1 \leq i \leq m$, and let $\mathcal{L}_* = \mathcal{L}(c)$. Suppose*

$$|y_i - c_i| \leq \delta \quad (1 \leq i \leq m), \qquad 0 \leq \delta < \tfrac{1}{2}. \tag{6.4}$$

*Then the points $y_i$ are strictly increasing and*

$$(1 + 2\delta)^{-2} \mathcal{L}_* \preceq \mathcal{L}(y) \preceq (1 - 2\delta)^{-2} \mathcal{L}_*. \tag{6.5}$$

*Here $A \preceq B$ means that $B - A$ is positive semidefinite. Furthermore,*

$$\|\mathcal{L}(y) - \mathcal{L}_*\|_{\mathrm{op}} \leq 16\delta \sum_{d=1}^{m-1} (d - 2\delta)^{-3}. \tag{6.6}$$

*Writing $0 = \lambda_1 < \lambda_2 \leq \cdots \leq \lambda_m$ for the ordered eigenvalues, one has*

$$\lambda_2(\mathcal{L}_*) \geq \frac{m}{(m-1)^2}, \qquad \|\mathcal{L}_*\|_{\mathrm{op}} \leq 4 \sum_{d=1}^{m-1} d^{-2}. \tag{6.7}$$

*Every nonzero eigenvalue of $\mathcal{L}(y)$ satisfies the corresponding multiplicative bounds in (6.5).*

*Proof.* If $i < j$ and $d = j - i$, then

$$d - 2\delta \le y_j - y_i \le d + 2\delta. \tag{6.8}$$

Since $d \ge 1$, these inequalities imply positivity of every adjacent gap and

$$(1 - 2\delta)d \le y_j - y_i \le (1 + 2\delta)d.$$

Insert the resulting inverse-square bounds into (6.3) to obtain (6.5).

On the positive interval between $d$ and $y_j - y_i$, the derivative of $r^{-2}$ has absolute value at most $2(d - 2\delta)^{-3}$. The mean-value theorem therefore gives

$$\left| W_{ij}(y) - \frac{1}{(j-i)^2} \right| \le \frac{4\delta}{(|j-i| - 2\delta)^3}.$$

The sum of the absolute values in a row of a Laplacian difference is at most twice the sum of its absolute weight differences. There are at most two indices at each label distance $d$. Consequently every absolute row sum is bounded by the right side of (6.6). For a real symmetric matrix its operator norm is at most its maximum absolute row sum, proving that estimate.

Every clock weight is at least $(m-1)^{-2}$. If $\sum_i v_i = 0$, the elementary identity

$$\sum_{i<j} (v_i - v_j)^2 = m \sum_i v_i^2$$

therefore proves the first bound in (6.7) by the Rayleigh quotient. The diagonal degree of any clock vertex is at most $2\sum_{d=1}^{m-1} d^{-2}$, so its absolute Laplacian row sum is at most twice that quantity. This proves the second bound. The eigenvalue comparisons follow from the minimum–maximum characterization of the ordered eigenvalues on the common orthogonal complement of constants. □

The physical normalization matters. If $y = \alpha x$, $\alpha > 0$, then

$$S(x) = \alpha S(y), \qquad \mathcal{L}(x) = \alpha^2 \mathcal{L}(y). \tag{6.9}$$

Thus convergence of $\mathcal{L}(y)$ to a finite matrix is consistent with divergence of the physical inverse-square weights if $\alpha \to \infty$. The convergence statement and its spatial scale must always be given together.

## 6.2 Complete root motion and the logarithmic energy

Let $H(t, z)$ be continuously differentiable in real $t$, holomorphic in $z$, real for real $t, z$, and satisfy

$$H_t = -H_{zz}. \tag{6.10}$$

On a real time interval $I$, let $x_1(t) < \cdots < x_m(t)$ be distinct simple real zeros, with their labels continued continuously. Implicit differentiation makes each label continuously differentiable. Put

$$P(t, z) = \prod_{i=1}^{m} (z - x_i(t)), \qquad H(t, z) = U(t, z)P(t, z).$$

The quotient is interpreted by removable continuation at the selected roots. It is nonzero near each of them. Define

$$b_i(t) = \frac{U_z(t, x_i(t))}{U(t, x_i(t))}. \tag{6.11}$$

These numbers are real. They retain every zero outside the selected finite population and every nonzero factor of the field. No infinite root sum is being truncated or reordered. A globally proper cluster provides one convenient way to construct $U$, but the calculations below require only these regular local factorizations along the roots.

**Proposition 6.2** (Complete finite-cluster dynamics)**.** *Under the preceding assumptions,*

$$x_i' = 2S_i(x) + 2b_i. \tag{6.12}$$

*For $i \neq j$, one has*

$$\frac{x_i' - x_j'}{x_i - x_j} = \frac{4}{(x_i - x_j)^2} - 2\sum_{k\neq i,j} \frac{1}{(x_i - x_k)(x_j - x_k)} + 2\frac{b_i - b_j}{x_i - x_j}, \tag{6.13}$$

$$W_{ij}(x)' = -2W_{ij}(x)\frac{x_i' - x_j'}{x_i - x_j}. \tag{6.14}$$

*Proof.* Differentiate $H(t, x_i(t)) = 0$, use (6.10), and divide by the nonzero $H_z(t, x_i)$:

$$x_i' = \frac{H_{zz}(t, x_i)}{H_z(t, x_i)}.$$

At a root, $H_z = UP_z$ and $H_{zz} = UP_{zz} + 2U_zP_z$. Writing $P = (z - x_i)\prod_{j\neq i}(z - x_j)$ gives

$$\frac{P_{zz}(x_i)}{P_z(x_i)} = 2\sum_{j\neq i} \frac{1}{x_i - x_j}.$$

This proves (6.12). Next,

$$S_i - S_j = \frac{2}{x_i - x_j} - (x_i - x_j)\sum_{k\neq i,j} \frac{1}{(x_i - x_k)(x_j - x_k)}.$$

Subtracting the two root equations proves (6.13). Ordinary differentiation of $(x_i - x_j)^{-2}$ proves (6.14). □

Define the squared Vandermonde and its negative logarithm by

$$\Delta(x) = \prod_{i<j}(x_j - x_i)^2, \qquad E(x) = -\log \Delta(x). \tag{6.15}$$

These are functions on the ordered simple-root chamber. The energy is singular at a collision; none of its differential formulas below is asserted at the collision itself.

**Lemma 6.3** (Pair identities and the forced energy balance)**.** *On that chamber,*

$$\nabla E = -2S, \qquad \nabla^2 E = 2\mathcal{L}(x), \qquad \|S(x)\|^2 = 2\sum_{i<j}(x_i - x_j)^{-2}. \tag{6.16}$$

*Along the complete heat motion,*

$$\begin{aligned} E' &= -4\|S\|^2 - 4S \cdot b \\ &= -8\sum_{i<j}(x_j - x_i)^{-2} - 4\sum_{i<j}\frac{b_j - b_i}{x_j - x_i}. \end{aligned} \tag{6.17}$$

*Equivalently,*

$$(\log \Delta)' = P_{\text{pair}} + B_{\text{ext}}, \qquad P_{\text{pair}} = 8\sum_{i<j}(x_j - x_i)^{-2}, \quad B_{\text{ext}} = 4\sum_{i<j}\frac{b_j - b_i}{x_j - x_i}. \tag{6.18}$$

*Proof.* Differentiating $-2\sum_{i<j}\log(x_j - x_i)$ gives the gradient and Hessian. In the expansion of $\sum_i S_i^2$, the diagonal summands give twice the inverse-square pair sum. For every unordered triple of distinct coordinates $a, b, c$, the remaining terms cancel by

$$\frac{1}{(a-b)(a-c)} + \frac{1}{(b-a)(b-c)} + \frac{1}{(c-a)(c-b)} = 0.$$

This proves the last identity in (6.16). Also, grouping the two terms belonging to each unordered pair gives

$$S \cdot b = \sum_{i<j} \frac{b_j - b_i}{x_j - x_i}.$$

The chain rule, $\nabla E = -2S$, and (6.12) now give (6.17); changing sign proves (6.18). □

The first term in (6.17) is negative, but the second has no sign under the stated assumptions. Positivity of the pair matrix does not remove it. In fact, the canonical application in Corollary 9.5 shows that the physical pair energy tends to positive infinity for every fixed late-time block. Consequently that energy cannot be nonincreasing on an entire late-time tail of the actual theta motion. This conclusion needs no differentiation of an asymptotic location error.

### 6.3 Normalized matrix evolution and a centered contraction bound

Let $\alpha(t) > 0$ be continuously differentiable and put $y_i = \alpha x_i$, $r = \alpha'/\alpha$. The normalized equation is

$$y' = 2\alpha^2 S(y) + f, \qquad f = ry + 2\alpha b. \tag{6.19}$$

In particular the correctly normalized weight evolution is

$$W_{ij}(y)' = -2W_{ij}(y)\Big[r + \frac{4\alpha^2}{(y_i - y_j)^2} - 2\alpha^2 \sum_{k\neq i,j} \frac{1}{(y_i - y_k)(y_j - y_k)} + 2\alpha\frac{b_i - b_j}{y_i - y_j}\Big]. \tag{6.20}$$

This formula follows either by differentiating $W_{ij}(y)$ directly in (6.19), or by differentiating $W_{ij}(y) = \alpha^{-2}W_{ij}(x)$. In particular the rescaling term $r$ is present in the normalized formula, and the physical formula (6.14) has no such term.

Fix an ordered reference $c \in \mathbb{R}^m$, independent of time. Let $P_0 v = v - m^{-1}(\sum_i v_i)\mathbf{1}$ be orthogonal projection onto the mean-zero subspace, and define

$$d = y - c, \quad d_0 = P_0 d, \qquad R = ry + 2\alpha b + 2\alpha^2 S(c). \tag{6.21}$$

The residual $R$ is a complete forced residual. Treating it as zero, or estimating only one of its summands, is an additional assertion about the field and is not part of the next theorem.

**Theorem 6.4** (Centered forced contraction)**.** *As long as $y$ and $c$ are strictly increasing,*

$$\frac{1}{2}\frac{d}{dt}\|d_0\|^2 = -2\alpha^2 \sum_{i<j} \frac{(d_i - d_j)^2}{(y_i - y_j)(c_i - c_j)} + d_0 \cdot P_0 R. \tag{6.22}$$

*Suppose in addition that $c_i = c_0 + i - 1$ and $|y_i - c_i| \le \delta(t) < 1/2$, with a continuous nonnegative allowance $\delta(t)$. Set*

$$w_0(t) = \frac{1}{(m-1)(m-1+2\delta(t))}, \qquad \lambda(t) = 2\alpha(t)^2 m w_0(t). \tag{6.23}$$

*Then, on every compact time interval on which these bounds hold,*

$$\|d_0(t)\| \le \exp\left(-\int_{t_0}^{t} \lambda(u)\,du\right)\|d_0(t_0)\| + \int_{t_0}^{t} \exp\left(-\int_{s}^{t} \lambda(u)\,du\right)\|P_0R(s)\|\,ds. \tag{6.24}$$

*The mean has its separate equation*

$$\frac{d}{dt}\left(\frac{1}{m}\sum_i y_i\right) = r\left(\frac{1}{m}\sum_i y_i\right) + \frac{2\alpha}{m}\sum_i b_i. \tag{6.25}$$

*Proof.* The identity

$$\frac{1}{y_i - y_j} - \frac{1}{c_i - c_j} = -\frac{d_i - d_j}{(y_i - y_j)(c_i - c_j)}$$

gives

$$S_i(y) - S_i(c) = -\sum_{j\neq i} \frac{d_i - d_j}{(y_i - y_j)(c_i - c_j)}.$$

Both vectors $S(y)$ and $S(c)$ have component sum zero. Subtract their contribution in (6.19), project onto the mean-zero subspace, and pair with $d_0$. Grouping unordered pairs proves (6.22), since $d_{0,i} - d_{0,j} = d_i - d_j$.

For clock $c$, both factors in each mixed denominator have the same sign. Their absolute product is at most $(m-1)(m-1+2\delta)$. The pair quadratic form is therefore at least

$$w_0 \sum_{i<j} (d_i - d_j)^2 = w_0 m \|d_0\|^2.$$

Cauchy–Schwarz in the forcing term gives, at times with $\|d_0\| > 0$,

$$\frac{d}{dt}\|d_0\| \le -\lambda\|d_0\| + \|P_0R\|.$$

At a zero of $d_0$, the same statement holds for the upper right Dini derivative, directly from the differentiable vector equation. Equivalently one may regularize the norm and pass to the limit. Multiplication by the integrating factor yields (6.24). Finally, summing (6.19) and using $\sum_i S_i = 0$ proves (6.25). □

This theorem identifies the estimate a proposed source-based relaxation argument must pay: the centered, complete residual $P_0R$. It is not a proof that this residual has a favorable sign or a sufficiently small norm for theta. Recovering $R$ from already-known root motion is an exact diagnostic, but supplies no new independent decay mechanism.

## 6.4 The unforced polynomial model and its Hermite limit

The polynomial case supplies a rigorous calibration of what deterministic pair repulsion does achieve. It also shows why regular late fibers alone do not exclude a multiple initial zero.

**Theorem 6.5** (Deterministic Hermite relaxation)**.** *Let $P(t,z)$ be a monic real polynomial of degree $m$ satisfying $P_t = -P_{zz}$. Suppose all its zeros are simple and real at some $t_0 > 0$. Then they remain*

*simple and real for every $t \geq t_0$. Let $x_1(t) < \cdots < x_m(t)$, put $u_i(t) = x_i(t)/\sqrt{2t}$, and set $s = \log t$. On the ordered chamber define*

$$V(u) = \frac{1}{4}\sum_i u_i^2 - \sum_{i<j} \log(u_j - u_i). \tag{6.26}$$

*This potential has a unique minimizer $h$; its coordinates are the ordered roots of the monic probabilists' Hermite polynomial $\mathrm{He}_m$, characterized by*

$$Q''(z) - zQ'(z) + mQ(z) = 0.$$

*The normalized motion and its estimates are*

$$\frac{du}{ds} = S(u) - \frac{1}{2}u = -\nabla V(u), \tag{6.27}$$

$$\frac{d}{ds}V(u(s)) = -\|\nabla V(u(s))\|^2, \tag{6.28}$$

$$\|u(t) - h\| \leq \left(\frac{t_0}{t}\right)^{1/2} \|u(t_0) - h\|, \tag{6.29}$$

$$V(u(t)) - V(h) \leq \frac{t_0}{t}\big(V(u(t_0)) - V(h)\big). \tag{6.30}$$

*Proof.* Initially the simple roots solve $x' = 2S(x)$, because the full polynomial has unit $U = 1$. Differentiating $x = \sqrt{2t}\,u$ and changing to $s = \log t$ gives (6.27). Direct differentiation of (6.26) gives

$$\nabla V = \frac{1}{2}u - S(u), \qquad \nabla^2 V = \frac{1}{2}I + \mathcal{L}(u) \succeq \frac{1}{2}I.$$

The potential tends to positive infinity at a finite collision. It also tends to positive infinity as $\|u\| \to \infty$: the quadratic term dominates the at-most-logarithmic negative contribution from each of the finitely many pair distances. Its sublevel sets are therefore compact subsets of the ordered chamber. Existence of a minimizer follows, and the displayed strict convexity on this convex chamber makes the minimizer unique.

At this minimizer, $2S_i(h) = h_i$. If $Q(z) = \prod_i (z - h_i)$, the same elementary root calculation used above yields $Q''(h_i)/Q'(h_i) = 2S_i(h) = h_i$. Thus $Q'' - zQ' + mQ$ vanishes at all $m$ distinct roots of $Q$. Its leading term cancels, so its degree is at most $m - 1$; it is identically zero. The monic polynomial satisfying this equation is unique by comparing coefficients in descending order. It is precisely $\mathrm{He}_m$. This also proves its real, simple root property here, rather than importing that property into the minimization argument.

The chain rule gives (6.28). The normalized trajectory cannot leave its initial compact sublevel set, so the smooth gradient equation extends for every finite forward $s$, with all gaps positive. Its coordinates reconstruct simple real zeros of the original polynomial: both obey the same locally unique root equation, and continuation gives their agreement. Hence no later finite collision or loss of reality occurs.

Strong convexity gives

$$(u - h) \cdot (\nabla V(u) - \nabla V(h)) \geq \frac{1}{2}\|u - h\|^2.$$

Differentiate squared distance along the gradient flow and integrate to obtain (6.29). Also strong convexity at $u$, with comparison point $h$, implies

$$V(u) - V(h) \leq \nabla V(u) \cdot (u - h) - \frac{1}{4}\|u - h\|^2 \leq \|\nabla V(u)\|^2.$$

Together with (6.28) this gives exponential decay of the potential gap in $s$, which is exactly (6.30) in physical time. □

For a direct event calibration, the finite heat series gives

$$e^{-t\partial_z^2} z^m = \sum_{k=0}^{\lfloor m/2 \rfloor} \frac{(-t)^k m!}{k!(m-2k)!} z^{m-2k} = (2t)^{m/2}\, \mathrm{He}_m\!\left(\frac{z}{\sqrt{2t}}\right). \tag{6.31}$$

Its normalized configuration is exactly $h$ for every $t > 0$, while its time-zero polynomial has one zero of multiplicity $m$. Perfect equilibrium of the normalized pair statistic therefore does not erase a past event. This is a statement about the explicit polynomial class, not an exclusion of an arithmetic argument for the canonical source. The deterministic minimizer in this theorem is one configuration; no random-matrix distribution has been produced.

The stochastic comparison is different: Dyson Brownian motion includes diffusion and, in its stationary Gaussian version, a confining drift. Its equilibrium is a probability distribution, whereas the gradient flow above converges to one minimizing configuration; see Joyner–Smilansky [3] for the stochastic equations.

# 7 Pair statistics and the inverse-square observable

A statistical description becomes useful for heat-flow topology only after its normalization, limiting procedure, and singular weights have been specified. We give a precise formulation and prove the relevant transfer statements. All populations in this section are finite, with labels retained when locations repeat.

Let $x_1, \dots, x_m \in \mathbb{R}$, let $\rho > 0$, and put $y_i = \rho x_i$. The factor $\rho$ is the chosen unfolding density. Define

$$\mu_m = \frac{1}{m} \sum_{i \neq j} \delta_{y_i - y_j}. \tag{7.1}$$

The sum is over ordered pairs of distinct labels. Thus a repeated location contributes an atom at zero even though the label diagonal is omitted. The sine-kernel pair density is

$$R_2(s) = 1 - \left(\frac{\sin \pi s}{\pi s}\right)^2, \qquad R_2(0) = 0. \tag{7.2}$$

A sine-kernel pair limit means vague convergence $\mu_m \to R_2(s)\, ds$, that is, convergence against every continuous compactly supported test function. Its meaning includes the choice of populations and of the density $\rho$.

Montgomery's original formulation concerns a growing height cutoff and unfolding by $\log T/(2\pi)$, with RH assumed in that paper; $T$ there is not a heat time. His discussion identifies the sine-kernel pair law and its random-matrix connection, and obtains density-one simplicity from the conjectured law [2, pp. 181–184]. In a heat problem one must separately prescribe the joint limits of heat time, height, and population. Fixed-rank late-time information alone does not specify a growing-population law.

If the points are distinct, define the real antisymmetric matrix

$$A_{ij} = \begin{cases} (y_i - y_j)^{-1}, & i \neq j, \\ 0, & i = j, \end{cases} \qquad Q_m = \frac{1}{m}\|A\|_F^2 = \int_{\mathbb{R}} \frac{d\mu_m(s)}{s^2}. \tag{7.3}$$

Here the integrand is assigned the value $+\infty$ at zero, so the last expression also defines an extended nonnegative quantity when labels repeat. The observable $Q_m$ is the ordered inverse-square pair energy per point. In physical coordinates,

$$8\sum_{i<j}\frac{1}{(x_i-x_j)^2}=4m\rho^2Q_m. \tag{7.4}$$

This is the pair term in the logarithmic derivative of the cluster discriminant. The complete heat formula also contains the exterior holomorphic factor; changing to matrix notation does not remove that factor. Moreover, at complex time the holomorphic weight $(z_i-z_j)^{-2}$ is distinct from the positive weight $|z_i-z_j|^{-2}$ in a Frobenius norm.

### 7.1 Exact clock and sine-kernel moments

**Proposition 7.1** (Two different inverse-square target values)**.** *For $m$ consecutive unit-spaced points,*

$$Q_m^{\mathrm{clock}}=\frac{2}{m}\sum_{d=1}^{m-1}\frac{m-d}{d^2}\longrightarrow\frac{\pi^2}{3}. \tag{7.5}$$

*The sine-kernel target instead has the finite moment*

$$\int_{\mathbb{R}}\frac{R_2(s)}{s^2}\,ds=\frac{2\pi^2}{3}. \tag{7.6}$$

*Proof.* There are $m-d$ unordered pairs at distance $d$, which proves the finite formula in (7.5). Rewriting it as

$$2\sum_{d=1}^{m-1}d^{-2}-\frac{2}{m}\sum_{d=1}^{m-1}d^{-1}$$

and using $\sum_{d\ge1}d^{-2}=\pi^2/6$ proves its limit. The latter identity follows, for example, by applying Parseval to the Fourier sine series of $x$ on $[-\pi,\pi]$, whose coefficients are $2(-1)^{d+1}/d$.

For the second claim, the elementary triangular Fourier identity is

$$\left(\frac{\sin\pi s}{\pi s}\right)^2=\int_{-1}^{1}(1-|u|)\cos(2\pi us)\,du. \tag{7.7}$$

It follows either by integrating the right side twice, or by taking the Fourier transform of the convolution of two interval indicators. Since $\int_{-1}^{1}(1-|u|)\,du=1$, the integrand in

$$\frac{R_2(s)}{s^2}=\int_{-1}^{1}(1-|u|)\frac{1-\cos(2\pi us)}{s^2}\,du$$

is nonnegative. Tonelli's theorem therefore applies without a prior integrability assumption. The elementary integral

$$\int_{\mathbb{R}}\frac{1-\cos(as)}{s^2}\,ds=\pi|a|$$

may be obtained by scaling and integration by parts, using the Dirichlet integral $\int_0^\infty\sin v/v\,dv=\pi/2$; the latter follows by first inserting $e^{-\varepsilon v}$, differentiating with respect to the sine frequency, and then letting $\varepsilon\downarrow0$. Consequently

$$\int_{\mathbb{R}}\frac{R_2(s)}{s^2}\,ds=2\pi^2\int_{-1}^{1}|u|(1-|u|)\,du=\frac{2\pi^2}{3}.$$

This calculation also proves finiteness. Directly, $R_2(s)=\pi^2s^2/3+O(s^4)$ at zero and $0\le R_2(s)\le1$ at infinity. □

The normalization difference in Proposition 7.1 is substantive. A clock and a sine-kernel population have different positive inverse-square energies; neither target energy is zero. If, for each fixed finite group of ranks, $L_t x_j(t)/\pi \to j - \frac{1}{2}$ as $t \to \infty$, then taking the heat limit first and the population limit second gives (7.5). A different simultaneous limit requires estimates uniform in its growing ranks and population.

## 7.2 Sparse exceptional points and the information in a pair law

**Lemma 7.2** (Sparse modifications preserve a local pair limit)**.** *For each $n$, let $y_1^{(n)}, \dots, y_n^{(n)}$ be a real labeled array. Suppose its pair measures (7.1) converge vaguely to a locally finite measure $\mu$. There is a choice of one old point in each array such that adjoining a second label at that same location preserves the vague limit. Alternatively, one can adjoin a distinct point at distance at most $n^{-2}$ from the chosen point, preserving the vague limit but ensuring*

$$Q_{n+1} \geq \frac{2n^4}{n+1}. \tag{7.8}$$

*If the original arrays are simple, the latter arrays can also be chosen simple.*

*Proof.* For every fixed positive integer $R$, dominate $\mathbf{1}_{[-R,R]}$ by a nonnegative continuous compactly supported function. Vague convergence gives

$$\#\{(i,j) : i \neq j,\ |y_i^{(n)} - y_j^{(n)}| \leq R\} = O_R(n).$$

A diagonal selection yields $R_n \to \infty$ sufficiently slowly that this count with $R = R_n$ is $o(n^2)$. For completeness, choose integers $N_k$ so large that for $n \geq N_k$ the count at radius $k$ is at most $n^2/k$, make $N_k$ increasing, and let $R_n$ be the largest admissible $k$ with $N_k \leq n$. Averaging over the first label gives an index $i_n$ with $o(n)$ neighbors inside radius $R_n$.

Adjoin a point $v_n = y_{i_n}^{(n)}$. For a continuous function $f$ supported in $[-R, R]$, the new pair integral equals

$$\frac{n}{n+1} \int f \, d\mu_n + \frac{1}{n+1} \sum_{j=1}^{n} [f(v_n - y_j^{(n)}) + f(y_j^{(n)} - v_n)]. \tag{7.9}$$

The first term tends to $\int f \, d\mu$. The sum in the second term has $o(n)$ nonzero summands, each bounded by $2\|f\|_\infty$, so it tends to zero after division by $n+1$. The new repeated pair contributes the atom $2\delta_0/(n+1)$, which is invisible in the limit.

For the alternative, choose $0 < \varepsilon_n \leq n^{-2}$ so that $v_n = y_{i_n}^{(n)} + \varepsilon_n$ is distinct from every old point. Only finitely many choices are forbidden. For each fixed $R$, its neighbors within $R$ are old neighbors of $y_{i_n}^{(n)}$ within $R+1$, so the proof using (7.9) is unchanged. The two ordered terms associated with $y_{i_n}^{(n)}$ give

$$Q_{n+1} \geq \frac{2}{(n+1)\varepsilon_n^2} \geq \frac{2n^4}{n+1}.$$

No existing points were moved, which proves the simplicity assertion. □

Thus weak pair convergence measures the bulk while admitting sparse exceptional pairs. This is a statement about the information content of a pair law, independent of whether a particular array can be realized as a zero set of a specified heat solution.

**Corollary 7.3** (Density-one simplicity from a sine-kernel pair law)**.** *If $\mu_n \to R_2(s)\,ds$, and a distinct location $a$ in the $n$th array has multiplicity $m_a$, then*

$$\frac{1}{n}\sum_a m_a(m_a-1) \longrightarrow 0. \tag{7.10}$$

*In particular, the fraction of labels lying at repeated locations tends to zero.*

*Proof.* The left side is $\mu_n(\{0\})$. For each $\delta > 0$ choose a continuous function $g_\delta$ with $0 \le g_\delta \le 1$, $g_\delta(0) = 1$, and support in $[-\delta, \delta]$. Then

$$\limsup_{n\to\infty} \mu_n(\{0\}) \le \int g_\delta(s) R_2(s)\,ds \le \int_{-\delta}^{\delta} R_2(s)\,ds.$$

The final bound tends to zero with $\delta$. At a repeated location $m_a \le m_a(m_a-1)$, proving the last assertion. □

### 7.3 Transfer of the singular inverse-square weight

**Proposition 7.4** (Singular-moment transfer)**.** *Suppose $\mu_n \to R_2(s)\,ds$ vaguely and*

$$\lim_{\delta\downarrow 0} \limsup_{n\to\infty} \int_{|s|<\delta} \frac{d\mu_n(s)}{s^2} = 0, \tag{7.11}$$

$$\lim_{R\to\infty} \limsup_{n\to\infty} \int_{|s|>R} \frac{d\mu_n(s)}{s^2} = 0. \tag{7.12}$$

*Then $Q_n \to 2\pi^2/3$.*

*Proof.* Fix $0 < \delta < R$. On the compact annulus $\delta \le |s| \le R$ the weight is bounded, and its only discontinuities when extended by zero are at four points of zero limiting measure. Continuous upper and lower cutoffs, or the usual weak-convergence criterion for bounded functions with a null discontinuity set, give

$$\int_{\delta\le|s|\le R} s^{-2}\,d\mu_n \longrightarrow \int_{\delta\le|s|\le R} s^{-2} R_2(s)\,ds.$$

The two nonnegative omitted terms are uniformly negligible in the successive limits by (7.11)–(7.12). The corresponding limiting terms vanish by Proposition 7.1. Removing the cutoffs proves the assertion. □

Conditions (7.11) and (7.12) identify the exact additional information needed by this moment transfer. A zero-gap atom makes the first integral infinite; that condition therefore cannot be assumed as an innocuous consequence of a weak pair law in an argument intended to establish simplicity. A regularized measurable observable is

$$Q_{n,\varepsilon} = \int_{\mathbb{R}} \frac{d\mu_n(s)}{s^2+\varepsilon^2}, \qquad \varepsilon > 0,$$

but its uniform behavior as $\varepsilon \downarrow 0$ is an additional estimate. Conversely, a genuinely proved finite bound $Q_m \le C$ gives

$$\min_{i\ne j} |y_i - y_j| \ge \sqrt{\frac{2}{mC}}, \tag{7.13}$$

because any one pair contributes two terms to $mQ_m$.

### 7.4 A finite-population error threshold

The following quantitative counting implication specifies how a sufficiently sharp pair estimate can establish more than density-one simplicity.

**Proposition 7.5** (Shrinking-tent test)**.** *Let $f_\delta(s) = (1 - |s|/\delta)_+$ for $\delta > 0$. Suppose a real labeled population of size $m$ has a proved enclosure*

$$\left| \int f_\delta \, d\mu_m - \int_{\mathbb{R}} f_\delta(s) R_2(s) \, ds \right| \le e_m(\delta). \tag{7.14}$$

*If*

$$\frac{\pi^2 \delta^3}{18} + e_m(\delta) < \frac{2}{m}, \tag{7.15}$$

*all locations are simple. If the right side in (7.15) is replaced by $1/m$, every two distinct labels are separated by more than $\delta/2$.*

*Proof.* By (7.7) and $1 - \cos v \le v^2/2$,

$$R_2(s) \le 2\pi^2 s^2 \int_{-1}^{1} u^2 (1 - |u|) \, du = \frac{\pi^2 s^2}{3}.$$

Since

$$\int_{-\delta}^{\delta} (1 - |s|/\delta) s^2 \, ds = 2 \left( \frac{\delta^3}{3} - \frac{\delta^3}{4} \right) = \frac{\delta^3}{6},$$

we obtain

$$\int f_\delta(s) R_2(s) \, ds \le \frac{\pi^2 \delta^3}{18}. \tag{7.16}$$

Any repeated location contributes at least two ordered terms of value $f_\delta(0) = 1$, hence $\int f_\delta \, d\mu_m \ge 2/m$. Equations (7.14)–(7.16) contradict this under (7.15). Likewise a pair at distance at most $\delta/2$ contributes two terms each at least $1/2$, giving the threshold $1/m$. □

For example, $\delta_m = o(m^{-1/3})$ and $e_m(\delta_m) = o(m^{-1})$ suffice for eventual simplicity of these finite populations. This is a conditional quantitative test: neither the shrinking-test uniformity nor its error bound follows from vague PCC. The threshold states an analytic task rather than an estimate for the canonical theta source.

## 8 Physical projective caps and their phase

We next derive a boundary formula that identifies the information beyond root positions needed for the physical Chern–Hopf integral. Let $\tau = t + is$ and $z = x + iy$. Let $H(\tau, z)$ be holomorphic and put

$$G = (H, H_z), \qquad \mathcal{A} = \frac{\operatorname{Im}(G^* dG)}{|G|^2}, \qquad \mathcal{C} = \frac{\mathcal{A} \wedge d\mathcal{A}}{4\pi^2}. \tag{8.1}$$

These are the physical jet and its normalized connection and three-form; in particular, the derivative component has not been rescaled relative to the value component.

**Lemma 8.1** (Normalization of the boundary three-form)**.** *If $W$ is a compact oriented four-dimensional domain with piecewise $C^1$ boundary, $G = (H, H_z)$ is nonzero on that boundary, and $\partial W$ has the outward orientation, then (with the complex orientation when $W \subset \mathbb{C}^2$)*

$$\int_{\partial W} \mathcal{C} = \deg\left(\frac{G}{|G|} : \partial W \longrightarrow S^3\right).$$

*For a proper product window satisfying Theorem 4.10, this is the discriminant charge in (4.16).*

*Proof.* On the unit sphere in $\mathbb{C}^2$, let $\alpha = \operatorname{Im}(\overline{\zeta}_1 d\zeta_1 + \overline{\zeta}_2 d\zeta_2)$. At $(1, 0)$, an oriented orthonormal tangent basis is $(\partial_{y_1}, \partial_{x_2}, \partial_{y_2})$, and $\alpha \wedge d\alpha$ evaluates to 2. Unitary invariance and the transitive action of $U(2)$ therefore give $\alpha \wedge d\alpha = 2\, d\operatorname{Vol}_{S^3}$ everywhere. The area of the unit three-sphere is $2\pi^2$ (by polar integration of the four-dimensional Gaussian), so this form divided by $4\pi^2$ has integral one. The physical connection is the pullback of $\alpha$ under $G/|G|$, since differentiating the real norm adds no imaginary part. The degree formula for a normalized volume form proves the first claim. Proposition 4.11 proves the last. □

This identifies the total complete-boundary integral. It does not identify an individual scalar discriminant edge integral from Proposition 4.13 with an individual physical three-dimensional cap; the latter is evaluated next.

Fix real numbers $t_0 < T$ and a rectangle

$$D = \{\tau : t_0 \le \operatorname{Re}\tau \le T,\ |\operatorname{Im}\tau| \le Y\}, \qquad Y > 0,$$

and a number $\varepsilon > 0$. Assume there is a holomorphic map $Z(\tau, q)$ on a neighborhood of $D \times \{|q| \le \varepsilon\}$ such that

$$H_z(\tau, Z(\tau, q)) \neq 0, \qquad \frac{H(\tau, Z(\tau, q))}{H_z(\tau, Z(\tau, q))} = q, \tag{8.2}$$

and $q \mapsto Z(\tau, q)$ is one-to-one for each $\tau$. The inverse derivative is nonzero, because differentiating the second identity gives $(H/H_z)_z Z_q = 1$. Thus $(\tau, q) \mapsto (\tau, Z(\tau, q))$ is an embedded tube, with the same orientation as $dt \wedge ds \wedge d\operatorname{Re} q \wedge d\operatorname{Im} q$. Indeed the spatial Jacobian is $|Z_q|^2 > 0$. Its central branch $Z(\tau, 0)$ is a simple root.

**Proposition 8.2** (Exact phase formula for a physical cap)**.** *Under (8.2), set*

$$h(\tau, q) = H_z(\tau, Z(\tau, q)), \qquad k_\varepsilon = \frac{\varepsilon^2}{1 + \varepsilon^2}.$$

*Choose a continuous phase $\phi = \operatorname{Im}\log h$ on the parameter tube. For the forward real-time cap at $t = b$, oriented by increasing $s$ followed by the complex $q$ orientation,*

$$I_b = \int_{t=b} \mathcal{C} = \frac{k_\varepsilon}{2\pi}\left[\phi(b + iY, 0) - \phi(b - iY, 0)\right]. \tag{8.3}$$

*The projective-circle wall $|q| = \varepsilon$ has identically zero pullback of $d\mathcal{A}$ and hence of $\mathcal{C}$.*

*Proof.* The parameter domain is contractible and $h$ is nonvanishing, so it admits a logarithm. In these coordinates $G = h(q, 1)$, whence direct substitution in (8.1) gives

$$\mathcal{A} = d\phi + a(q), \qquad a(q) = \frac{\operatorname{Im}(\overline{q}\, dq)}{1 + |q|^2}, \qquad d\mathcal{A} = \omega(q) := \frac{2\, du \wedge dv}{(1 + u^2 + v^2)^2}, \quad q = u + iv. \tag{8.4}$$

Both $a$ and $\omega$ use only the two $q$ coordinates, so $a \wedge \omega = 0$. Therefore

$$\mathcal{C} = \frac{d\phi \wedge \omega}{4\pi^2}. \tag{8.5}$$

On a fixed real-time cap, only $\phi_s\, ds$ in $d\phi$ contributes. For each $s$, $q \mapsto \phi_s(b + is, q)$ is harmonic: it is the imaginary part of the $s$ derivative of the holomorphic logarithm of $h$. Angular averaging on every centered circle gives its central value. Since $\omega$ has radial density,

$$\int_{|q|\le\varepsilon} \phi_s(b + is, q)\, \omega(q) = \phi_s(b + is, 0) \int_{|q|\le\varepsilon} \omega(q).$$

The curvature mass is

$$\int_{|q|\le\varepsilon} \omega = \int_0^{2\pi} \int_0^{\varepsilon} \frac{2r}{(1 + r^2)^2}\, dr\, d\theta = 2\pi k_\varepsilon.$$

Integrating in $s$ proves (8.3). On the wall, $q = \varepsilon e^{i\theta}$ depends on only one real parameter. Thus the two-form $\omega$ pulls back to zero, proving the wall assertion. □

Equation (8.5) retains the common scalar phase. Replacing $G = h(q, 1)$ by its projective representative $(q, 1)$ would give $a \wedge da = 0$ and would remove the very term measured by (8.3). It is therefore essential to carry the phase of $H_z$ along the continued root. This observation concerns the normalized jet; arbitrary multiplication of $H$ by a time-dependent scalar is not asserted to preserve the heat equation.

## 8.1 The complete regular tube

**Proposition 8.3** (The imaginary-time faces close the cap identity)**.** *In the setting of Proposition 8.2, let $J_+$ and $J_-$ be the outward oriented integrals on $s = Y$ and $s = -Y$, respectively. Then*

$$J_+ = -\frac{k_\varepsilon}{2\pi}[\phi(T + iY, 0) - \phi(t_0 + iY, 0)], \tag{8.6}$$

$$J_- = \frac{k_\varepsilon}{2\pi}[\phi(T - iY, 0) - \phi(t_0 - iY, 0)], \tag{8.7}$$

$$I_T - I_{t_0} + J_+ + J_- = 0. \tag{8.8}$$

*If $H$ and the inverse branch have real type, there is a continuous choice*

$$\theta(t, Y) = \operatorname{Im} \log \frac{h(t + iY, 0)}{h(t, 0)}$$

*for which*

$$I_t = \frac{k_\varepsilon}{\pi}\theta(t, Y), \qquad J_+ = J_- = \frac{k_\varepsilon}{2\pi}[\theta(t_0, Y) - \theta(T, Y)]. \tag{8.9}$$

*Proof.* The ambient orientation is $dt \wedge ds \wedge du \wedge dv$. Contraction with the upper outward normal $\partial_s$ gives $-dt \wedge du \wedge dv$, whereas the lower outward normal $-\partial_s$ gives $+dt \wedge du \wedge dv$. Apply the same harmonic radial averaging as in Proposition 8.2, now to $\phi_t$, and integrate in $t$. This yields (8.6)–(8.7); substitution of (8.3) proves (8.8) by cancellation of its four endpoint values. Equivalently, $d\mathcal{C} = \omega \wedge \omega/(4\pi^2) = 0$ on this regular chart, so Stokes gives the same identity, with the zero wall already paid.

For the last claim, $h(t, 0)$ is nonzero real on the real segment, and therefore has constant sign. The logarithm after division by that sign can be chosen real on the real segment. Conjugation then gives opposite imaginary parts at $t + iY$ and $t - iY$. Inserting these two values in the preceding formulas gives (8.9). □

This proposition makes a useful distinction: an exactly vanishing projective wall is compatible with nonzero real-time caps, because the imaginary-time faces provide the remaining flux. The assertion requires a fixed projective radius on each tube. If the radius varies with time, the wall pulls back $dr \wedge d\theta$ terms and requires a separate calculation. A tube satisfying (8.2) contains no joint zero of $(H, H_z)$; attaching it to an event-containing region does not remove that region's other boundary pieces.

## 8.2 Exact heat examples

We give two explicit calculations within the backward heat convention $H_\tau = -H_{zz}$. They isolate the respective roles of root regularity and physical phase without assuming a statistical model.

**Proposition 8.4** (A heat event followed by simple clock asymptotics)**.** *The entire real-type field*

$$H(\tau, z) = e^\tau \cos z + 1 \tag{8.10}$$

*satisfies the backward heat equation. At $(\tau, z) = (0, \pi)$ it has an ordinary double spatial zero and local complex degree one for the map $(H, H_z)$. At every positive real time all its spatial zeros are real and simple, and its late real zero pattern converges to a clock of spacing $\pi$.*

*Proof.* Direct differentiation gives $H_\tau = e^\tau \cos z = -H_{zz}$. At $(0, \pi)$, $H = H_z = 0$, $H_{zz} = 1$, and the complex Jacobian is

$$\frac{\partial(H, H_z)}{\partial(\tau, z)}(0, \pi) = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Its nonzero determinant makes the zero of this holomorphic map simple, with local degree one. For $t > 0$, the equation is $\cos z = -e^{-t} \in (-1, 0)$. If $z = x + iy$ with $y \neq 0$ and $\cos z$ is real, then $\sin x = 0$, so $|\cos z| = \cosh y > 1$, a contradiction. Thus all zeros are real, and $\sin z \neq 0$ at them, proving simplicity. The two roots in $(0, 2\pi)$ are

$$x_\pm(t) = \pi \pm \arccos(e^{-t}).$$

They tend to $\pi/2$ and $3\pi/2$; translation by multiples of $2\pi$ gives the clock limit. For $t < 0$ the same local pair is $\pi \pm i \operatorname{arcosh}(e^{-t})$, displaying explicitly the transition through the event. □

The source of (8.10) is a positive atomic cosine measure. The example proves that late simplicity and clock relaxation are compatible with earlier event charge in the general heat class; it makes no assertion about an event for the canonical theta source.

**Proposition 8.5** (Identical tracked roots with different cap phases)**.** *For $a > 0$, the two exact real-type heat fields*

$$H_1(\tau, z) = z, \qquad H_2(\tau, z) = e^{a^2\tau} \frac{\sin(az)}{a} \tag{8.11}$$

*have the same simple tracked root $z = 0$ at every complex time. For any $0 < \varepsilon < 1/a$, their root inverse disks are respectively*

$$Z_1(\tau, q) = q, \qquad Z_2(\tau, q) = \frac{1}{a} \arctan(aq), \tag{8.12}$$

*and their physical cap integrals at $t = T$, $-Y \le s \le Y$, are*

$$I_{T,1} = 0, \qquad I_{T,2} = \frac{\varepsilon^2}{1 + \varepsilon^2} \frac{a^2 Y}{\pi}. \tag{8.13}$$

*Proof.* Both heat equations follow by differentiation. Their derivatives at the tracked root are 1 and $e^{a^2\tau}$, so that root is simple. The projective quotients are $q = z$ and $q = \tan(az)/a$. The principal branch of $\arctan(aq)$ is holomorphic on $|q| < 1/a$; it is one-to-one there because its tangent recovers $aq$, and its derivative is $(1 + a^2q^2)^{-1} \neq 0$. Thus both inverse disks satisfy (8.2). On the second disk,

$$h_2(\tau, q) = \frac{e^{a^2\tau}}{\sqrt{1 + a^2q^2}},$$

with the square root equal to one at $q = 0$. The central phases are therefore 0 and $a^2s$. Formula (8.3) gives (8.13). □

The last example identifies a precise information deficit of a finite tracked population: its positions do not specify the physical phase of the derivative along the root. The second field has exterior roots; no claim is made about nonuniqueness from an entire global zero set. Together, the pair-energy results and the cap formula provide a concrete route for future estimates: control the weighted pair statistic in a stated limit, retain its complete exterior contribution, and then estimate the complex-time derivative phase and all faces of the chosen boundary.

# 9 Complete-source theta clock limits and growing windows

This section supplies a self-contained canonical application of the finite matrix theorem. The proof uses the entire positive theta source, an explicit elementary concentration bound, and a real Fourier comparison. It proves locations, simplicity, rank identification and matrix convergence at large real heat time, including the growing windows specified below. It does not differentiate a root location remainder. A separate, stronger differentiated saddle theorem would be needed for pointwise velocity asymptotics; those asymptotics are not used here.

## 9.1 The source and its elementary bounds

Use the fixed normalization

$$H(t, z) = \int_0^\infty e^{tu^2}\Phi(u)\cos(zu)\,du, \qquad \Phi(u) = \sum_{n=1}^{\infty}(2\pi^2n^4e^{9u} - 3\pi n^2e^{5u})e^{-\pi n^2e^{4u}}. \tag{9.1}$$

In particular the complete summands are positive for $u \geq 0$. Writing

$$\Phi(u) = 2\pi^2e^{9u-\pi e^{4u}}\mathfrak{b}(u), \tag{9.2}$$

$$\mathfrak{b}(u) = \sum_{n=1}^{\infty}\left(n^4 - \frac{3n^2}{2\pi}e^{-4u}\right)e^{-\pi(n^2-1)e^{4u}}, \tag{9.3}$$

we have the global bounds

$$0 < \mathfrak{b}_- := 1 - \frac{3}{2\pi} \leq \mathfrak{b}(u) \leq \mathfrak{b}_+ := \frac{1}{1 - 16e^{-3\pi}} < \infty. \tag{9.4}$$

Indeed the $n = 1$ term supplies the lower bound, and every term is positive. For the upper bound discard only the negative summand in this upper estimate and use $n \leq 2^{n-1}$, $n^2 - 1 \geq 3(n - 1)$, and $e^{4u} \geq 1$. The resulting series is bounded by the geometric series with ratio $16e^{-3\pi} < 1$. No term is discarded from the field in (9.1) itself.

The upper bound in (9.4) implies convergence of (9.1) and all its parameter derivatives uniformly on compact subsets of $\mathbb{C}^2$. In fact every such derivative is bounded in absolute value by a constant times

$$\int_0^\infty (1+u)^j e^{Au^2+Bu+9u-\pi e^{4u}}\,du < \infty$$

for suitable finite $A, B, j$. Thus $H$ is holomorphic in both variables and satisfies $H_t = -H_{zz}$ by differentiation under the integral. For real $t$, its normalizing value $H(t,0)$ is strictly positive.

For $L \geq 2$, define

$$t = t(L) = \frac{4\pi e^{4L} - 9}{2L}, \qquad \kappa = 16\pi e^{4L} - 2t = 8tL - 2t + 36. \tag{9.5}$$

The function $t(L)$ is strictly increasing, because

$$t'(L) = \frac{4\pi e^{4L}(4L-1)+9}{2L^2} > 0.$$

It maps $[2,\infty)$ onto $[t_2,\infty)$, where $t_2 = (4\pi e^8 - 9)/4$. We use its inverse to define $L = L(t)$ for every $t \geq t_2$. Set

$$q_t(u) = tu^2 + 9u - \pi e^{4u}. \tag{9.6}$$

Then $q_t'(L) = 0$ and $q_t''(L) = -\kappa < 0$. The identities (9.5) imply

$$\kappa \geq 36, \qquad \frac{t}{\kappa} \leq \frac{1}{8L-2} \leq \frac{1}{14} < \frac{1}{4e}, \qquad \pi e^{4L} \leq \frac{\kappa}{8}. \tag{9.7}$$

For the last inequality, writing $A = \pi e^{4L}$, one has $t = (4A-9)/(2L) < A$ for $L \geq 2$, and hence $\kappa = 16A - 2t > 14A$.

## 9.2 An explicit first-moment concentration theorem

Let $U$ denote a nonnegative random variable with the complete probability law

$$d\mu_t(u) = \frac{e^{tu^2}\Phi(u)}{H(t,0)}\,du. \tag{9.8}$$

This notation is only a convenient integral normalization; the zero configuration itself is deterministic. Put $c_* = (4e)^{-1}$.

**Theorem 9.1** (Complete elementary concentration bound)**.** *For every $t \geq t_2$, with $L, \kappa$ as in (9.5),*

$$\mathbb{E}_t|U - L| \leq M_1(t) := \frac{e\mathfrak{b}_+}{\mathfrak{b}_-}\left\{\frac{1}{c_*\sqrt{\kappa}} + \frac{L^2\sqrt{\kappa}}{2}e^{-c_*\kappa/16}\right\}. \tag{9.9}$$

*In particular*

$$M_1(t) = O(\kappa^{-1/2}), \qquad L \sim \tfrac{1}{4}\log t, \qquad \kappa \sim 8tL \quad (t \to \infty). \tag{9.10}$$

*The bound includes every theta mode and the entire half-line of integration.*

*Proof.* Write $A = \pi e^{4L}$, $v = u - L$, and $a = 1/4$. The saddle equation gives the exact identity

$$q_t(L+v) - q_t(L) = tv^2 - A(e^{4v} - 1 - 4v). \tag{9.11}$$

For $v \geq 0$, Taylor's formula with positive remainder gives $e^{4v} - 1 - 4v \geq 8v^2$, so

$$q_t(L+v) - q_t(L) \leq -\frac{\kappa}{2}v^2. \tag{9.12}$$

For $-a \leq v \leq 0$, the integral second-order Taylor remainder instead gives $e^{4v} - 1 - 4v \geq 8e^{-1}v^2$. Since $8A = (\kappa + 2t)/2$,

$$t - 8A/e = -\frac{\kappa}{2e} + t(1 - e^{-1}) \leq -\frac{\kappa}{4e} = -c_*\kappa$$

by (9.7). Combining both sides yields

$$q_t(L+v) - q_t(L) \leq -c_*\kappa v^2 \qquad (-a \leq v < \infty). \tag{9.13}$$

We must also pay the whole far-left interval $0 \leq u \leq L - a$. The second derivative $q_t''(u) = 2t - 16\pi e^{4u}$ is strictly decreasing. Any local maximum of $q_t$ must be on the descending part of $q_t'$, where its only zero is the already identified point $L$. Thus $q_t$ has no interior local maximum in $[0, L-a]$; its maximum there is at an endpoint. Moreover,

$$q_t(L-a) = t\left((L - \tfrac{1}{4})^2 - \frac{L}{2e}\right) + 9\left(L - \tfrac{1}{4} - \frac{1}{4e}\right) > 0,$$

where both brackets are positive for $L \geq 2$. Since $q_t(0) = -\pi < 0$, the right endpoint is the larger one. Equation (9.13) at $v = -a$ consequently gives the complete far-left bound

$$q_t(u) - q_t(L) \leq -\frac{c_*\kappa}{16} \qquad (0 \leq u \leq L - \tfrac{1}{4}). \tag{9.14}$$

For a normalization lower bound, use $|v| \leq 1/(2\sqrt{\kappa})$. This interval lies in $[-a, a]$ and in the physical half-line after translation by $L$. The elementary upper Taylor bound

$$0 \leq e^{4v} - 1 - 4v \leq 8e^{4|v|}v^2$$

and $A \leq \kappa/8$ show that

$$q_t(L+v) - q_t(L) \geq -\kappa e^{4|v|}v^2 \geq -e/4 > -1.$$

Here $4|v| \leq 2/\sqrt{\kappa} < 1$. Therefore

$$Z_t := \int_0^\infty e^{q_t(u) - q_t(L)}\mathfrak{b}(u)\,du \geq \frac{\mathfrak{b}_- e^{-1}}{\sqrt{\kappa}}. \tag{9.15}$$

Using (9.13), (9.14), and $\mathfrak{b} \leq \mathfrak{b}_+$, the first absolute-moment numerator is at most

$$\begin{aligned}
&\int_0^\infty |u - L| e^{q_t(u) - q_t(L)}\mathfrak{b}(u)\,du \\
&\quad \leq \mathfrak{b}_+ \left\{ \int_{\mathbb{R}} |v| e^{-c_*\kappa v^2}\,dv + e^{-c_*\kappa/16} \int_{-L}^{-1/4} |v|\,dv \right\} \\
&\quad \leq \mathfrak{b}_+ \left\{ \frac{1}{c_*\kappa} + \frac{L^2}{2} e^{-c_*\kappa/16} \right\}.
\end{aligned}$$

Division by (9.15) proves (9.9).

Finally,

$$t = \frac{2\pi e^{4L}}{L} - \frac{9}{2L}, \qquad \kappa = 4\pi e^{4L}(4 - L^{-1}) + 9L^{-1}.$$

These formulas prove $L \sim (\log t)/4$ and $\kappa \sim 8tL$. They also give $\kappa \geq 14\pi e^{4L}$ for $L \geq 2$, and therefore $L^2\kappa e^{-c_*\kappa/16} \to 0$. The second term in (9.9) is consequently $o(\kappa^{-1/2})$, proving its stated order. □

### 9.3 The Fourier transfer and the first finitely many roots

The following simple transfer isolates exactly which information from concentration is used. It does not assume real-rootedness of the entire function.

**Lemma 9.2** (Real Fourier concentration transfer)**.** *Let $U \geq 0$ have a probability law with finite first moment, let $L > 0$, and let $m_1 = \mathbb{E}|U - L|$. For real $z$ put $F(z) = \mathbb{E}\cos(zU/L)$. Then for every $Z > 0$,*

$$\sup_{|z|\leq Z} |F(z) - \cos z| \leq \frac{Zm_1}{L}, \qquad \sup_{|z|\leq Z} |F'(z) + \sin z| \leq \frac{(1+Z)m_1}{L}. \tag{9.16}$$

*Proof.* Write $U/L = 1 + V/L$, with $V = U - L$. Since cosine is 1-Lipschitz on the real line,

$$|\cos(zU/L) - \cos z| \leq |z||V|/L.$$

The finite first moment permits differentiation under the expectation. For the derivative use

$$\begin{aligned}\left|\frac{U}{L}\sin(zU/L) - \sin z\right| &\leq \frac{|V|}{L}|\sin(zU/L)| + |\sin(zU/L) - \sin z| \\ &\leq (1 + |z|)|V|/L.\end{aligned}$$

Taking expectations proves both bounds. □

**Theorem 9.3** (Canonical fixed-rank clock and matrix limit)**.** *Fix $K \geq 1$ and $0 < \beta < \pi/4$, and set*

$$s_j = (j - \tfrac{1}{2})\pi, \qquad Z = s_K + \beta, \qquad \epsilon_0(t) = \frac{ZM_1(t)}{L}, \quad \epsilon_1(t) = \frac{(1+Z)M_1(t)}{L}. \tag{9.17}$$

*Suppose $t \geq t_2$ and*

$$\epsilon_0(t) < \sin\beta, \qquad \epsilon_1(t) < \cos\beta. \tag{9.18}$$

*Then the first $K$ positive real zeros of $H(t,\cdot)$ exist, are simple, and satisfy*

$$|Lx_j(t) - s_j| \leq e(t) := \arcsin\epsilon_0(t) < \beta, \qquad 1 \leq j \leq K. \tag{9.19}$$

*All these finite tests hold for every sufficiently large $t$, and*

$$e(t) = O_K((L\sqrt{\kappa})^{-1}). \tag{9.20}$$

*No RH assumption is needed for these assertions about the indicated late-time positive roots.*

*In particular, fix $\ell \geq 0$, $m \geq 2$, $K = \ell + m$, and put*

$$y_i = \frac{L}{\pi}x_{\ell+i}(t), \qquad c_i = \ell + i - \tfrac{1}{2}, \qquad \delta(t) = e(t)/\pi. \tag{9.21}$$

*The finite matrix bounds (6.5) and (6.6) hold with these data. Consequently*

$$\|\mathcal{L}(y) - \mathcal{L}_*\|_{\mathrm{op}} = O_K((L\sqrt{\kappa})^{-1}), \tag{9.22}$$

$$\left\|\mathcal{L}(x_{\ell+1},\dots,x_{\ell+m}) - \frac{L^2}{\pi^2}\mathcal{L}_*\right\|_{\mathrm{op}} = O_K(L/\sqrt{\kappa}). \tag{9.23}$$

*The second error tends to zero, while the leading physical matrix grows like $L^2$.*

*Proof.* Normalize the actual field by

$$F_t(z) = \frac{H(t, z/L)}{H(t, 0)} = \mathbb{E}_t \cos(zU/L).$$

Theorem 9.1 and Lemma 9.2 give

$$|F_t(z) - \cos z| \le \epsilon_0, \qquad |F_t'(z) + \sin z| \le \epsilon_1 \quad (|z| \le Z).$$

On the interval $[s_j - \beta, s_j + \beta]$, the endpoint cosine values have opposite signs and absolute value $\sin\beta$. The first strict test in (9.18) preserves those signs for $F_t$. On that interval the function $\sin z$ has one sign and absolute value at least $\cos\beta$. The second strict test therefore gives a nonzero derivative with fixed sign for $F_t$. There is exactly one simple zero in each interval. On the complement of these intervals in $[0, Z]$, one has $|\cos z| \ge \sin\beta$, so the first test excludes every other positive real zero there. This proves the rank statement.

At the zero in the $j$-th interval, $|\cos z| = |\sin(z - s_j)| \le \epsilon_0$. Since $|z - s_j| < \beta < \pi/2$, inversion of sine gives (9.19). The concentration theorem shows that both $\epsilon_0$ and $\epsilon_1$ tend to zero, with size $O_K((L\sqrt{\kappa})^{-1})$; hence the tests eventually hold and so does (9.20).

For the selected block, $|y_i - c_i| \le \delta < 1/4$. Apply Theorem 6.1. Since $m$ is fixed, its finite sum in (6.6) is bounded uniformly for $0 \le \delta < 1/4$, proving (9.22). The physical rate follows by multiplying that bound by $L^2/\pi^2$, using (6.9). □

The proof establishes simplicity on these large-time real root blocks. It does not establish simplicity at time zero, nor does it identify a hypothetical initial multiple root with one of these labels without additional global continuation hypotheses. The asymptotic statements above fix the maximum rank $K$. The explicit finite tests also permit the following uniform spatial consequence.

**Corollary 9.4** (Growing clock windows and dimension-independent matrix error)**.** *Fix $0 < \beta < \pi/4$ and let $K = K(t) \ge 1$ be integer valued. At each time define $Z, \epsilon_0, \epsilon_1$ by (9.17) with this value of $K$. Whenever (9.18) holds, all first $K$ positive real roots are simple and obey (9.19) simultaneously. For every contiguous block of $m \ge 2$ roots ending by rank $K$, set $y_i = Lx_{\ell+i}/\pi$, $c_i = \ell + i - 1/2$, and $\delta = \pi^{-1}\arcsin\epsilon_0 < 1/4$. Then*

$$\|\mathcal{L}(y) - \mathcal{L}(c)\|_{\mathrm{op}} \le \frac{16\zeta(3)}{(1-2\delta)^3}\delta \le 128\zeta(3)\delta. \tag{9.24}$$

*In particular, the sufficient condition*

$$K(t) = o(L(t)\sqrt{\kappa(t)}) \tag{9.25}$$

*makes the finite tests hold eventually and gives, uniformly over all such blocks,*

$$\max_{j \le K} |Lx_j - s_j| = O\left(\frac{K}{L\sqrt{\kappa}}\right) = o(1), \tag{9.26}$$

$$\|\mathcal{L}(y) - \mathcal{L}(c)\|_{\mathrm{op}} = O\left(\frac{K}{L\sqrt{\kappa}}\right), \tag{9.27}$$

$$\left\| \mathcal{L}(x_{\ell+1}, \dots, x_{\ell+m}) - \frac{L^2}{\pi^2}\mathcal{L}(c) \right\|_{\mathrm{op}} = O\left(\frac{KL}{\sqrt{\kappa}}\right). \tag{9.28}$$

*The constants do not depend on $K$, $m$ or the block's starting rank. The stronger condition $K = o(\sqrt{\kappa}/L)$ is sufficient for the absolute physical-matrix error to tend to zero.*

*Proof.* The sign, monotonicity and complementary-interval arguments in Theorem 9.3 are finite-time implications with explicit $K$ dependence, so apply to $K(t)$ at each time. For $d \geq 1$, $d-2\delta \geq (1-2\delta)d$, and therefore

$$\sum_{d=1}^{m-1} (d-2\delta)^{-3} \leq (1-2\delta)^{-3} \sum_{d=1}^{\infty} d^{-3} = (1-2\delta)^{-3}\zeta(3).$$

Insert this in (6.6); $\delta < 1/4$ makes $(1-2\delta)^{-3} \leq 8$. This proves (9.24) without a population factor. Since $Z \leq \pi K$ and $1+Z \leq (1+\pi)K$, the $K$-independent bound $M_1 = O(\kappa^{-1/2})$ proves the eventual tests and the first two rates. Multiplication by $L^2/\pi^2$ gives the physical rate. Finally $L \to \infty$, so the stronger condition also implies (9.25). □

At each time the compared matrices have the same dimension, which may grow; no operator limit between unspecified varying spaces is asserted. The corollary is spatial and requires no time regularity of $K(t)$. Changing the population in time creates entry and exit terms. In particular it does not extend the next fixed-population discriminant identity by substituting $m(t)$, nor does it estimate the complete centered exterior residual or yield a pair-correlation law.

### 9.4 Cumulative exterior cancellation without velocity asymptotics

The location theorem also proves a useful dynamical consequence. It is weaker than a pointwise estimate for $(\log \Delta)'$, but requires no differentiated saddle analysis and exposes the necessary exterior payment in a fully analytic way.

**Corollary 9.5** (Physical energy growth and cumulative cancellation)**.** *Fix the block from Theorem 9.3 and choose $T_0$ large enough that its strict finite tests hold for every $t \geq T_0$. The root labels are real analytic on this time interval. Define $\Delta(t)$, $P_{\mathrm{pair}}(t)$, and $B_{\mathrm{ext}}(t)$ by (6.15) and (6.18), with the complete local unit of the actual theta field. Put*

$$A_m = \sum_{d=1}^{m-1} \frac{m-d}{d^2}, \qquad C_m = m(m-1)\log\pi + 2\sum_{d=1}^{m-1}(m-d)\log d. \tag{9.29}$$

*Then*

$$\log \Delta(t) = -m(m-1)\log L(t) + C_m + o(1), \tag{9.30}$$

$$P_{\mathrm{pair}}(t) = \frac{8A_m}{\pi^2} L(t)^2 + O_K\left(\frac{L(t)}{\sqrt{\kappa(t)}}\right), \tag{9.31}$$

$$\int_{T_0}^{T} P_{\mathrm{pair}}(t)\,dt \sim \frac{8A_m}{\pi^2} T L(T)^2, \tag{9.32}$$

$$\int_{T_0}^{T} B_{\mathrm{ext}}(t)\,dt = -\int_{T_0}^{T} P_{\mathrm{pair}}(t)\,dt - m(m-1)\log L(T) + O(1). \tag{9.33}$$

*In particular $E(t) = -\log\Delta(t) \to +\infty$, and*

$$\frac{\displaystyle\int_{T_0}^{T} B_{\mathrm{ext}}(t)\,dt}{\displaystyle\int_{T_0}^{T} P_{\mathrm{pair}}(t)\,dt} \longrightarrow -1. \tag{9.34}$$

*These are cumulative statements. They assert no sign for the pointwise remainder obtained by differentiating (9.30).*

*Proof.* The field is holomorphic in both variables, and each selected zero is simple. The implicit-function theorem gives local analytic labels; the unique rank identification in Theorem 9.3 makes these labels agree on overlapping intervals.

For $i < j$, the root enclosure gives

$$x_{\ell+j} - x_{\ell+i} = \frac{\pi}{L}\left((j-i) + O_K((L\sqrt{\kappa})^{-1})\right).$$

There are finitely many pairs. Taking logarithms and summing proves (9.30), with $C_m$ as displayed. The inverse-square expansion, or equivalently the trace of (9.23), gives (9.31).

The relation $L(t) \sim (\log t)/4$ implies

$$\int_{T_0}^{T} L(t)^2\,dt \sim \frac{1}{16} T(\log T)^2 \sim TL(T)^2.$$

To see this directly, fix any relative error, use the asymptotic comparison with $(\log t)^2/16$ beyond a fixed threshold, and integrate the upper and lower bounds; the finite initial interval is negligible. The elementary antiderivative $\int (\log t)^2 dt = t((\log t)^2 - 2\log t + 2)$ then proves the assertion. Applying the same comparison argument to $P_{\mathrm{pair}}(t)/L(t)^2 \to 8A_m/\pi^2$ proves (9.32).

Finally the exact identity (6.18) integrates to

$$\int_{T_0}^{T} B_{\mathrm{ext}}(t)\,dt = \log\Delta(T) - \log\Delta(T_0) - \int_{T_0}^{T} P_{\mathrm{pair}}(t)\,dt.$$

Substitute (9.30) to obtain (9.33). The logarithmic remainder is negligible relative to the positive quantity in (9.32), proving (9.34). The same log-discriminant formula proves $E(t) \to +\infty$. □

The complete finite-cluster unit therefore cannot be omitted when the inverse-square matrix is used dynamically. Its cumulative contribution cancels an order-$TL(T)^2$ internal pair contribution, leaving an order-$\log L(T)$ change in the log discriminant. This conclusion is specific to the proved fixed-rank clock regime and the complete field. It is neither an exclusion of other functionals nor a proof of large-population random-matrix statistics.

## 9.5 Limits of the statistical interpretation

For the fixed block, the normalized ordered pair differences tend to the integers $j - i$. For example its empirical measure

$$\nu_{m,t} = \frac{1}{m(m-1)} \sum_{i \neq j} \delta_{y_j(t) - y_i(t)}$$

converges weakly to

$$\frac{1}{m(m-1)} \sum_{i \neq j} \delta_{j-i}.$$

This follows immediately by evaluating any bounded continuous test function on the finitely many converging differences. It is an atomic clock limit. It is not a claim about a smooth limiting pair-correlation density, a random matrix ensemble, or a high-height limit at fixed heat time. Nor does fixed-rank convergence imply convergence uniformly over all roots. Those populations and limiting operations have to be specified and estimated separately.

Finally, the full-source proof above exploits definite analytic properties of the canonical source, but its conclusion alone is not a simplicity discriminator at an earlier time. The exact Hermite birth in (6.31) already exhibits simple late roots and a multiple initial zero. In the present theta

regime the late physical gaps shrink, while the normalized gaps converge to positive values. Neither feature supplies a vanishing complex-time boundary integral or pays an unmeasured closure face. Such a boundary consumer still needs the complete field, its exterior unit and amplitude, its chosen topology, and every term in its own transport identity.

## 10 Consequences and further analytic questions

The localized energy theorem identifies an exact analytic representative of the full zero count. Its validity through multiple zeros is a strength for transport: the count is not lost at a collision. The same property means that it cannot by itself exclude a collision. The regularized measure records the divisor in a weak limit, with an explicit field-error and localization cost. These estimates can be used on any domain for which the boundary amplitude and the required local units are controlled.

For real heat evolution, the complete cap difference counts births only after wall crossings and endpoint conventions are included. For complex time, the discriminant gives the appropriate collision residue. The cover's topology, the energy of $2\,dw$, the reciprocal differential $dz/w$, and the physical jet phase each encode a different part of this accounting. The exact examples show why finiteness or regularity of one cannot be substituted for a vanishing statement about another.

The pair matrix supplies a quantitative geometric object with a precise heat law. Its theta clock limit is analytically controlled, including the full source tail, uniformly through $K = o(L\sqrt{\kappa})$ in the normalization of Corollary 9.4. An absolute physical-matrix error tending to zero follows in the smaller window $K = o(\sqrt{\kappa}/L)$. These spatial estimates do not give a growing-population dynamical law or the statistical transfer conditions of Section 7. A dynamical improvement would require an estimate for the complete exterior forcing, rather than an unforced repulsion model. A cap estimate would additionally require the root-derivative phase and the complete closing faces of Section 8.

Theorem 3.6 settles the fixed-disk $C^2$ weighted-divisor question with explicit constants and a sharp exponent. Its complete cluster and boundary data remain essential. Concrete further analytic questions are whether a coupled source estimate controls the centered exterior residual in a useful growing regime; whether comparable divisor estimates persist under a specified weaker information contract or uniformly through controlled merging clusters; and whether a full physical boundary admits a signed estimate stronger than the identities alone. The present results settle none of those questions by a numerical experiment or by a change of terminology. They provide explicit formulas, errors and examples against which such estimates can be tested.

## A A supporting certified finite disk

This appendix illustrates the analytic identity using existing numerical data; it is not used in any theorem above. The recorded finite-fiber calculation [9] uses the canonical identity

$$H_0(z) = \frac{1}{8}\xi\left(\frac{1+iz}{2}\right)$$

and outward-rounded complex enclosures. It certifies exactly two zeros in $\Omega = \{|z-35| < 8\}$ and the full boundary margin

$$\min_{|z-35|=8} |H_0(z)| > 6.6759 \times 10^{-7}. \tag{A.1}$$

The count is certified by a complete boundary winding calculation; the supporting record also isolates two simple real roots. Decimal approximations to the roots alone are not the certificate.

If these recorded count and boundary-margin enclosures are accepted, Theorem 2.2 gives the exact consequence

$$W_{<\varepsilon}(H_0;\Omega) = 8\pi\varepsilon, \qquad 0 < \varepsilon \le 6 \times 10^{-7}. \tag{A.2}$$

For example the value at $\varepsilon = 10^{-7}$ is exactly $8\pi \times 10^{-7}$. This is the localized derivative energy, not the unweighted reciprocal energy $2\int |H_0|^{-1}\, dA$. The equality uses the already certified count; it does not independently certify it.

There is also a source-defined positive continuation disk. With the complete theta source of Section 9, put

$$M_2 = \int_0^\infty u^2 e^{u^2+8u}\Phi(u)\, du, \qquad h = \min\{1, 3 \times 10^{-7}/M_2\}.$$

The source decay proves that $M_2$ is finite. On $|\tau| \le 1$ and $|z - 35| = 8$, differentiating the defining integral gives $|H_\tau(\tau, z)| \le M_2$. Integration along the time segment therefore gives $|H(\tau, z) - H_0(z)| \le 3 \times 10^{-7}$ for $|\tau| \le h$. The boundary margin is then greater than $3.6759 \times 10^{-7}$, and Rouché's theorem preserves count two. Consequently (A.2) holds throughout $|\tau| \le h$ for $0 < \varepsilon \le 3 \times 10^{-7}$. No simplicity of every continued fiber is required, and no useful numerical size for $h$ is claimed without evaluating the moment.

No new numerical replay or acquisition is claimed in this manuscript. The analytic proofs, including the separate large-time theta result, do not depend on (A.1).

## Acknowledgements, provenance and status

This manuscript was drafted for Edward Baker with substantial assistance from OpenAI's Codex, recorded model `gpt-6-astra`, reasoning effort *ultra*. LLM assistance was used for derivation, checking, exposition and document preparation. Separate agents of the same model reviewed the transport, localized-energy and theta-concentration arguments. These are internal checks, not independent specialist or human validation. The divisor-stability and growing-window continuation used the recorded model `gpt-6-astra`, reasoning effort *max*; its proof audit is a same-assistant self-review, not a separate review.

The work consolidates and extends the author's research archive on physical jet geometry, relative Chern–Hopf charges, reciprocal Hodge energy and late heat caps [8]. In particular the real wall/birth identity and the complex event charge were established in earlier notes; their proofs are included here. The exact low-value count is a refinement of the same area formula used for the earlier full-energy count bound. Classical ingredients are identified in the text, and no priority claim is made. The manuscript is a working analytic treatment with the stated hypotheses and limitations. Its supporting numerical appendix imports an existing certificate rather than reporting a new computation. Detailed internal reviews and a source-to-result crosswalk are retained with the manuscript's research records.

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[9] E. Baker, *Actual regular theta fiber comparison*, research note prepared with substantial LLM assistance, 22 October 2026; associated outward-ball certificate and review in the research archive. Local source: `papers/investigations-4/09_reciprocal_hodge_energy/notes/04_ACTUAL_REGULAR_THETA_FIBER_COMPARISON_20261022.md`. This is the supporting numerical input of Appendix A only.